

SEALS User Guide

Spatial Economic Allocation Landscape Simulator

Justin Andrew Johnson

2026-08-01

Table of contents

1	What SEALS is	4
1.1	How to read this guide	4
1.2	Where SEALS ends and the platform begins	5
1.3	Status and citation	5
2	SEALS Environment Setup	6
2.1	1. Install Miniforge/Conda	6
2.2	2. Create and activate an environment	6
2.3	3a. User install	6
2.4	3b. Developer install from a GitHub clone	7
2.5	4. Add other devstack repos (optional)	8
3	Quickstart	9
3.1	1. Install	9
3.2	2. Run it	9
3.3	3. Where the output goes	10
3.4	4. Make it your own	10
3.5	Next	11
4	First SEALS run walkthrough	12
4.1	Getting set up	12
4.2	Explore the SEALS code	12
4.3	Run files	13
4.4	Part 1: the task tree	13
4.5	Part 2: run_project	13
4.6	Part 3: the __main__ guard	14
4.7	The p object	14
4.8	Where the project directory goes	14
5	Scenario definitions	16
5.1	The scenarios CSV	16
5.2	Scenario types	16
5.3	Automatically downloading data	16
6	Running it	18
6.1	Run the model	18

6.2	Find the output	18
6.3	Next	18
7	scenarios.csv Format (v1)	19
7.1	Column reference	19
7.2	Example	21
7.3	Where the file lives	21
8	Release Notes	22
8.1	SEALS v2.0.0 (2026-07-30)	22
I	Coupling SEALS to other models	24
9	Magpie	25
9.1	Installation	25
10	To run gams	26
11	Applying SEALS to downscale MAgPIE outputs	27
11.1	Overview	27
11.2	Create MAgPIE outputs for SEALS by using output scripts	27
11.3	Setting up SEALS to downscale MAgPIE outputs	32
II	Worked examples	36
12	Worked example: Kenya DEMETRA-SEALS-InVEST	37
12.1	Overview	37
12.2	Installation	38
12.3	Data Setup	40
12.4	Testing SEALS	41
12.5	Kenya DEMETRA Scenario	42
12.6	Understanding the Task Tree	43
12.7	Task-by-Task Walkthrough	44
12.8	Final Outputs	63
12.9	Additional Resources	64

1 What SEALS is

SEALS, the Spatial Economic Allocation Landscape Simulator, is a land-use change model that downscales predictions of land-use change from aggregate (regional or coarse-gridded) inputs to a finer resolution, typically 10–300 meters.

Its primary comparative advantage is fast computation. It downscales to 300 meters globally — roughly 8.4 billion grid cells — in about an hour on a laptop. It is designed with parallelization in mind and so scales well to high-performance, cloud, or distributed computing. Performance-critical functions are written in C++ (via Cython); everything else is Python.

SEALS is one of three software products maintained by NatCap TEEMs (The Earth-Economy Modellers):

product	what it is
SEALS	this model — the downscaler
GTAP-InVEST	the coupled economy–ecosystem model that uses SEALS for its land-use step
Earth-Economy Devstack	the shared platform: hazelbean, ProjectFlow, and the conventions all three follow

1.1 How to read this guide

Start at [Environment Setup](#) and [Quickstart](#) — together they take you from nothing to a downscaled map. Then:

- [Walkthrough](#) — a guided first run, and the anatomy of a run file. Read this before you start a project of your own.
- [scenarios.csv format](#) — the column reference for the file that defines what SEALS runs.
- [MAGPIE](#) and the [MAGPIE tutorial](#) — coupling SEALS to an external land-use model, which is also the general pattern for coupling it to any coarse projection source.

This guide is rendered both as a website and as a single PDF ([download](#)). If you are reading the PDF, links to the other two products point at the published site.

1.2 Where SEALS ends and the platform begins

SEALS is a model. The machinery that runs it — project directories, task trees, skip-if-already-computed re-runs, automatic base-data download — is **ProjectFlow**, which lives in `hazelbean` and is shared across all three products. This guide covers SEALS and says only as much about ProjectFlow as you need to run it.

For the platform itself, see the Earth-Economy Devstack documentation:

- [Conventions](#) — the run-file spec, naming rules, and directory layout
- [Project complexity](#) — the three axes that describe how a run file is organized, and why
- [ProjectFlow](#) — the task-tree engine
- [Installation](#) — devstack-wide install options, including the developer install

1.3 Status and citation

SEALS is under active development and changes rapidly. Pull the latest `seals` before reporting a problem.

The model was introduced alongside GTAP-InVEST in [Johnson et al. 2023, PNAS](#). All code is available under a permissive open-source license.

2 SEALS Environment Setup

Two paths: a **user install** if you just want to run SEALS, or a **developer install** from a GitHub clone if you want to edit SEALS (or hazelbean) and have your edits take effect immediately.

SEALS is distributed as **sealsmodel** and imported as **seals**. It requires hazelbean 2.0.0 or later.

2.1 1. Install Miniforge/Conda

If not already installed, download and install [Miniforge](#).

2.2 2. Create and activate an environment

```
conda create -n <your_env> python=3.10
conda activate <your_env>
```

Pick your own environment name — the devstack does not require a particular one. Always activate it before running SEALS scripts.

2.3 3a. User install

```
pip install hazelbean sealsmodel
```

Dependencies are installed automatically. This is enough to run `run_seals_standard.py` and to start your own project from it.

2.4 3b. Developer install from a GitHub clone

Use this if you cloned the repos and want your edits to take effect without reinstalling. SEALS compiles a Cython extension (`seals/seals_cython_functions.pyx`), so a **C/C++ compiler is required** — see [Compiling C/C++ Code](#) (`xcode-select --install` on macOS; Visual Studio Build Tools on Windows).

Clone into the devstack layout. Each repo lives in its own project folder under `~/Files/` (`C:\Users\<you>\Files\` on Windows). The layout matters: ProjectFlow infers where to put project directories from it, placing them at `~/Files/seals/projects/<project_name>/` — outside the repo, so outputs never land in your working tree.

```
mkdir -p ~/Files/hazelbean && cd ~/Files/hazelbean
git clone https://github.com/jandrewjohnson/hazelbean_dev.git

mkdir -p ~/Files/seals && cd ~/Files/seals
git clone https://github.com/jandrewjohnson/seals.git
```

If you already installed hazelbean from conda-forge, remove it first so the editable install is the one that gets imported. Use `conda remove`, not `pip uninstall` — `pip` leaves behind directories that still shadow the source:

```
conda remove hazelbean --force
```

This drops the conda-forge build but keeps its dependencies.

Make the editable installs, `hazelbean` first — SEALS imports it:

```
cd ~/Files/hazelbean/hazelbean_dev
pip install -e . --no-deps

cd ~/Files/seals/seals
pip install -e . --no-deps
```

`--no-deps` is deliberate: the dependencies came from conda in step 2, and letting `pip` resolve them again tends to replace conda's builds of the geospatial stack (GDAL and friends) with incompatible wheels.

Verify:

```
python -c "import hazelbean, seals; print(hazelbean.__file__); print(seals.__file__)"
```

Both paths should point inside `~/Files/...`, not into `site-packages`. If either points at `site-packages`, an installed copy is shadowing your clone — remove it and redo the editable install.

2.5 4. Add other devstack repos (optional)

The same clone-and-editable-install pattern works for the rest of the stack, each in its own ~/Files/<name>/ project folder: [gtap_invest_dev](#), [global_invest_dev](#), [gtappy_dev](#), [gep_dev](#).

See the [full installation guide](#) for the devstack-wide version of these steps.

3 Quickstart

Get SEALS running, then make it your own. For the model itself see [SEALS](#); for the full install options see [Installation Steps](#).

3.1 1. Install

SEALS is distributed as `sealsmodel` and imported as `seals`. It needs [hazelbean](#), which supplies `ProjectFlow`.

```
conda create -n <your_env> python=3.10
conda activate <your_env>
pip install hazelbean sealsmodel
```

Pick your own environment name. If you intend to edit SEALS rather than just run it, do a developer install from a clone instead — see [environment_setup.md](#) in the seals repo, which also covers the C/C++ compiler SEALS needs to build its Cython extension.

3.2 2. Run it

Two run files ship with SEALS, in `seals/seals/`:

file	what it runs
<code>run_seals_standard.py</code>	the standard pipeline: three scenarios, 2030 and 2050
<code>run_seals_standard_test.py</code>	the same pipeline pared down — baseline plus one BAU, one year, Rwanda

Start with the test:

```
conda activate <your_env>
cd seals/seals
python run_seals_standard_test.py
```

Required base data is downloaded at runtime from a public bucket; no credentials are needed for the default configuration.

3.3 3. Where the output goes

Not into the repo. Because the run file lives inside a cloned repo, ProjectFlow places the project directory just *outside* it:

```
~/Files/seals/projects/seals_standard_test/  
  input/           seeded from the repo's input_template/  
  intermediate/    one folder per task  
  output/
```

Run the same command a second time and it finishes almost immediately — every task is behind an existence check, so completed work is skipped. That is the main thing ProjectFlow buys you: interrupt a long run, fix one input, re-run, and only what is missing recomputes.

3.4 4. Make it your own

Copy the run file into your own project repo as `run_<project>.py` and change the two lines in its `__main__` block — that is the whole of the configuration:

```
if __name__ == '__main__':  
    p = hb.ProjectFlow(project_name='my_project', run_mode='check')  
    p.scenario_definitions_filename = 'my_project_scenarios.csv'  
  
    run_project(p)
```

`run_mode` controls how much prior work is reused: `'check'` (the default) resumes in a stable project dir; `'full'` timestamps a fresh one.

Put your scenarios CSV in an `input_template/` directory beside the run file. It is tracked in git, and ProjectFlow copies anything missing into the project's untracked `input/` on first run without ever overwriting your working copy. The columns are documented in [scenarios_format.md](#).

To run the same pipeline a second way — a smoke test, a different AOI — write a *second run file* that imports the first rather than copying it. `run_seals_standard_test.py` is the worked example, and the pattern is explained in [conventions](#).

3.5 Next

- [Walkthrough](#) — a guided tour with screenshots
- [Conventions](#) — the full run-file rules
- [Project complexity](#) — why run files are shaped this way

4 First SEALS run walkthrough

4.1 Getting set up

- Follow the [installation](#) page first, or the shorter [quickstart](#) if you just want to run the model
- Clone SEALS and Hazelbean into the devstack layout — each repo in its own project folder under ~/Files/:

- ~/Files/hazelbean/hazelbean_dev
 - ~/Files/seals/seals

- The layout is not cosmetic: ProjectFlow reads it to decide where project directories go
- You know the install worked if this prints two paths inside ~/Files/, not inside `site-packages`:

```
python -c "import hazelbean, seals; print(hazelbean.__file__); print(seals.__file__)"
```

4.2 Explore the SEALS code

- The repo root `seals/` holds more than the library — docs, tests, scripts
- The library itself is the `seals/seals` subdirectory. The nesting looks redundant but is how Python packaging works
- Inside it, `__init__.py` is what makes the directory importable as a package
- The modules are split by what they hold:

- `seals_main.py`, `seals_tasks.py`, `seals_generate_base_data.py` — task functions, the actual model logic
 - `seals_initialize_project.py` — the task-tree builders
 - `seals_utils.py` — helpers
 - `run_seals_standard.py` — the run file

4.3 Run files

- You do not run `seals_main.py` directly
- A **run file** configures a project and executes a task tree built from the library's task functions
- Two ship with SEALS, both in `seals/seals`:
 - `run_seals_standard.py` — three scenarios, 2030 and 2050
 - `run_seals_standard_test.py` — the same pipeline pared to a baseline, one BAU scenario, one year, Rwanda. Start here
- Open `run_seals_standard.py`. It is about 60 lines and has three parts

4.4 Part 1: the task tree

- `build_task_tree(p)` answers one question: *what is this pipeline?*
- It contains nothing but tree construction. For stock SEALS it is one line, delegating to the shared library builder
- Compose extra library subtrees or your own tasks here if your project diverges

```
def build_task_tree(p):  
    seals_initialize_project.build_standard_task_tree(p)
```

4.5 Part 2: run_project

- `run_project(p)` answers *how is it executed?* It takes only `p`
- The rule that decides what goes where:
 - **`run_project(p)` sets what no variant ever changes** — base data location, processing resolution
 - **the caller sets what a variant might** — project name, run mode, scenarios CSV
- When a constant starts varying, it moves one line up into the caller. There is no signature to edit and no default to keep in sync

```
def run_project(p):  
    p.run_in_parallel = 1          # before the tree: it has parallel iterators  
    build_task_tree(p)  
    p.skip_tasks(p.tasks_to_skip)  
  
    p.base_data_dir = os.path.join(p.user_dir, 'Files', 'base_data')
```

```

p.processing_resolution = 1.0

p.scenario_definitions_path = os.path.join(p.input_dir, p.scenario_definitions_filename)
seals_initialize_project.initialize_scenario_definitions(p)
seals_initialize_project.set_advanced_options(p)

p.execute()
return p

```

4.6 Part 3: the `__main__` guard

- This is where the project is configured — two lines of actual choice
- The guard is mandatory: importing a run file must never start a run. That is what lets a second run file import this pipeline instead of copying it

```

if __name__ == '__main__':
    p = hb.ProjectFlow(project_name='seals_standard', run_mode='check')
    p.scenario_definitions_filename = 'standard_scenarios.csv'

    run_project(p)

```

4.7 The `p` object

- `hb.ProjectFlow()` is a class — a recipe for an object. Calling it produces an object, which we assign to `p`
- `p` carries the project's **attributes**: `p.base_data_dir`, `p.scenario_definitions_filename`, and so on
- It also has **methods** that act on it: `p.add_task()`, `p.skip_tasks()`, `p.get_path()`, `p.execute()`
- Every task receives `p` and reads its configuration from it. That is how configuration set in `__main__` reaches code deep in the library

4.8 Where the project directory goes

- Not into the repo. Because the run file sits inside a cloned repo, `ProjectFlow` places the project dir just *outside* it, at `~/Files/seals/projects/<project_name>/`
- You do not compute this path yourself, and you should not save data in `seals/seals`
- `run_mode` decides how much prior work is reused:

- 'check' — reuse the stable dir, recompute only what is missing (the default)
- 'fresh_intermediate' — redo all computation, keep `input/` (test projects only)
- 'full' — a fresh timestamped dir; everything re-runs

5 Scenario definitions

5.1 The scenarios CSV

- The scenarios CSV specifies the runs you want. **Each row is one scenario**
- As the model iterates, it re-hydrates `p` from the row it is on — which is why scenario-varying values belong in the CSV and not in the run file
- The tracked copy lives in `input_template/` beside the run file. ProjectFlow copies anything missing into the project's `input/` on first run and never overwrites your working copy
- If the named file is nowhere to be found, SEALS generates a default so a first run still works

```
p.scenario_definitions_filename = 'standard_scenarios.csv'
```

5.2 Scenario types

- `scenario_type` is a **column in the CSV**, not something a run file sets
- `baseline` — the observed anchor; not downscaled, supplies base-year LULC
- `policy` — a trajectory to downscale, compared against a counterfactual
- `bau` — business-as-usual, being deprecated in favour of `baseline`; it still appears in shipped CSVs
- Full column reference: [scenarios_format.md](#)

5.3 Automatically downloading data

- Paths in the CSV are `ref_paths` — relative locations resolved by `p.get_path()`, which searches the project `input/` dir, then `base_data`, then a cloud bucket, downloading what is missing
- `p.base_data_dir` is where those downloads land. Point it somewhere with room for very large files; the same base data serves every project
- The directory must be named `base_data` to match the bucket convention
- The default bucket is public — no credentials needed

```
p.base_data_dir = os.path.join(p.user_dir, 'Files', 'base_data')
```

6 Running it

6.1 Run the model

```
conda activate <your_env>
cd seals/seals
python run_seals_standard_test.py
```

- On startup SEALS prints the **task tree** it is about to compute
- To understand the model in depth, read the task functions behind those names
- Run the same command a second time: it finishes almost immediately, because every task is behind an existence check and completed work is skipped

6.2 Find the output

- Go to `~/Files/seals/projects/seals_standard_test/intermediate/`
- There is one directory per task in the tree
- The downscaled maps are in `stitched_lulc_simplified_scenarios/`, for example:

`lulc_esa_seals7_ssp2_rcp45_luh2-message_bau_2030.tif`

- Open it in QGIS alongside the base-year LULC and compare — that is your high-resolution land-use change projection

6.3 Next

- [Quickstart](#) — the condensed version of this page
- [Conventions](#) — the full run-file rules
- [Project complexity](#) — why run files are shaped this way

7 scenarios.csv Format (v1)

The scenarios CSV defines your SEALS scenarios. Each row is one scenario. The tracked copies that ship with SEALS are [seals/input_template/standard_scenarios.csv](#) (three scenarios) and `standard_scenarios_test.csv` (a pared two-row version); this page documents their columns.

This is the v1 schema — the one the code reads today. A v2 schema is specified in [scenario_definitions.qmd](#) and no code implements it yet. v2 fixes several structural problems in v1: it replaces `baseline_reference_label` / `comparison_counterfactual_labels` / per-model time columns with a `observed_reference_label` / `compared_to` / `depends_on` split and a single canonical clock, and it allows hindcasting. Write v1 for now; expect a migration when the hazelbean parser lands.

7.1 Column reference

Values in the Example column are taken from the shipped `standard_scenarios.csv`.

Column	Description	Example
<code>scenario_label</code>	Unique scenario identifier	<code>baseline_luh2-message,</code> <code>ssp2_rcp45_luh2-</code> <code>message_bau</code>
<code>scenario_type</code>	<code>baseline</code> , <code>bau</code> , or <code>policy</code> — see the note below	<code>baseline</code>
<code>aoi</code>	Area of interest: <code>global</code> , an ISO3 code, or a path to a vector	<code>RWA</code>
<code>exogenous_label</code>	Exogenous driver set	<code>baseline</code> , <code>ssp2</code>
<code>climate_label</code>	Climate scenario	<code>rcp45</code>
<code>model_label</code>	Coarse-projection model source	<code>luh2-message</code>
<code>counterfactual_label</code>	Counterfactual this row represents (blank on the baseline row)	<code>bau</code> , <code>bau_shift</code>
<code>years</code>	Space-separated projection years	<code>2030 2050</code>

Column	Description	Example
<code>baseline_reference_label</code>	Scenario label of the baseline row this one is measured against (blank on the baseline row)	<code>baseline_luh2-message</code>
<code>base_years</code>	Observed/base year(s)	2017
<code>key_base_year</code>	The base year used for LULC	2017
<code>comparison_counterfactual_label</code>	Scenario label this row is compared against in reporting	<code>ssp2_rcp45_luh2- message_bau</code>
<code>time_dim_adjustment</code>	Offset applied to the source time dimension	<code>add2015</code>
<code>coarse_projections_input_path</code>	Coarse projections NetCDF	<code>luh2/raw_data/rcp45_ssp2/multiple- states_..._2015- 2100.nc</code>
<code>lulc_src_label</code>	Fine LULC source	<code>esa</code>
<code>lulc_simplification_label</code>	Fine LULC class scheme	<code>seals7</code>
<code>lulc_correspondence_path</code>	Fine class correspondence	<code>seals/default_inputs/esa_seals7_corre</code>
<code>coarse_src_label</code>	Coarse data source	<code>luh2-14</code>
<code>coarse_simplification_label</code>	Coarse class scheme	<code>seals7</code>
<code>coarse_correspondence_path</code>	Coarse class correspondence	<code>seals/default_inputs/luh2- 14_seals7_correspondence.csv</code>
<code>lc_class_varname</code>	Land-class variable in the NetCDF	<code>all_variables</code>
<code>dimensions</code>	Dimensions to read	<code>time</code>
<code>calibration_parameters_source</code>	Downscaling coefficients	<code>seals/default_inputs/default_global_c</code>
<code>base_year_lulc_path</code>	Base-year LULC raster	<code>lulc/esa/lulc_esa_2017.tif</code>
<code>regional_projections_input_path</code>	Regional projections CSV (optional; blank if coarse-only)	<code>seals/default_inputs/rwa_bdi_regional</code>
<code>regions_vector_path</code>	Regions vector	<code>cartographic/ee/ee_r264_correspondenc</code>
<code>regions_column_label</code>	Region column in that vector	<code>ee_r264_label</code>

Paths are **ref_paths**: relative locations resolved by `p.get_path()`, which searches the project `input/` dir, then `base_data`, then the cloud bucket, downloading if needed. Always write them with **forward slashes**, on every platform.

7.1.1 A note on scenario_type

Three values appear in current files:

- **baseline** — the observed anchor. Not downscaled; it supplies base-year LULC and calibration inputs.

- **bau** — a business-as-usual trajectory. **Deprecated in favour of baseline.** The migration is in progress and unevenly applied, so **bau** still appears in shipped CSVs and is still read by the code. Do not rely on it for new work, but do not be surprised to see it.
- **policy** — a policy trajectory, downscaled and typically compared against a BAU row via `comparison_counterfactual_labels`.

7.2 Example

The shipped `standard_scenarios.csv`: a Rwanda baseline, a BAU trajectory, and a policy row that shifts the BAU using regional projections.

```
scenario_label,scenario_type,aoi,exogenous_label,climate_label,model_label,counterfactual_label,
baseline_luh2-message,baseline,RWA,baseline,rcp45,luh2-message,,2017,,2017,2017,,add2015,luh2-
ssp2_rcp45_luh2-message_bau,bau,RWA,ssp2,rcp45,luh2-message,bau,2030 2050,baseline_luh2-messa
ssp2_rcp45_luh2-message_bau_shift,policy,RWA,ssp2,rcp45,luh2-message,bau_shift,2030 2050,bas
```

Known wart: the shipped CSVs currently write `base_year_lulc_path` as `lulc\esa\lulc_esa_2017.tif`, with backslashes. The devstack rule is forward slashes everywhere, on every platform; these should be normalized.

7.3 Where the file lives

Put your scenarios CSV in an `input_template/` directory beside your run file. It is tracked in git; ProjectFlow copies anything missing into the project's untracked `input/` on first run and never overwrites the working copy. Name it in your run file's `__main__` block:

```
p.scenario_definitions_filename = 'my_project_scenarios.csv'
```

If the named file is absent from both `input/` and `base_data`, SEALS generates a default one in the project's `input/` so a first run still works — edit that copy, or better, add your own to `input_template/`.

8 Release Notes

Release history for SEALS. Versions continue this repository’s own tags (v1.2.0 preceded v2.0.0).

For the shared platform beneath SEALS — hazelbean, ProjectFlow and the conventions — see the [Earth-Economy Devstack release notes](#).

8.1 SEALS v2.0.0 (2026-07-30)

Run-file spec: three complexity axes, and `run_project(p)`.

- A run file’s complexity is now described by three *independent* axes rather than one ladder. **Configuration** (how a run varies) keeps the numbered levels 1–4: one task with inline constants; a task tree; + a scenarios CSV; + a parameters CSV. **Code layout** (where the code lives) runs single file → split (<project>_tasks.py, _functions.py, _utils.py, _initialize_project.py) → library package. **Ownership** (who owns the code you run) runs self-contained → devstack developer → downstream user. Moving along one axis never requires moving along another, so “*level*” *unqualified always means the configuration axis*.
- `run_project` now takes only `p`. The caller builds and configures the ProjectFlow in the `__main__` guard; `run_project(p)` only runs it. The rule that replaces a signature full of keyword defaults: **`run_project(p)` sets what no variant ever changes; the caller sets what a variant might**. A constant that starts varying moves one line up instead of becoming a new keyword argument, so the entry point stops growing as a project accumulates definition CSVs. Cost: bare `run_project()` no longer works, and a variant wrapper is four lines rather than three.
- `<project>_utils.py` is defined as a **promotion queue** into hazelbean — science-unaware helpers used by more than one project belong upstream — while `<project>_functions.py` holds the model’s domain knowledge and is terminal.
- `examples/` reorganized: `run_templates/` (copy-me) and `run_templates_annotated/` (the same code with the reasoning written out), plus `input_research_scripts/` for real run files and the raw scripts they replace. Two new templates hold configuration fixed at level 4 and move one other axis each: `run_template_split_layout.py` and `run_template_downstream_user.py`.

- The six numbered sections of *project_complexity.qmd* are renamed to “Stage N, the old way” / “Stage N, redone with ProjectFlow” so they no longer collide with the configuration levels. They are a historical narrative, not the spec.
- All 46 *_old_spec.py archives are deleted across the stack. Git history is the archive for pre-conversion forms. The empty docs/run_file.qmd stub is removed; examples/index.qmd is the landing page for “what should my run file look like?”.

Part I

Coupling SEALS to other models

9 Magpie

9.1 Installation

`git clone -b develop https://github.com/magpiemodel/magpie.git magpie_develop`

Has input and output folders in gitignore

MAGpie has a config folder

default.cfg

is something huge, so use a start script which takes lines from default and overwrites.

Then in the non-ignored starts folder, has a `test_runs.R`

10 To run gams

see `start/start_functions.R`

11 Applying SEALS to downscale MAgPIE outputs

11.1 Overview

The Model of Agricultural Production and Its Impact on the Environment (MAgPIE) is a global food and land system modelling framework. It uses a wide range of spatially-explicit biophysical and socioeconomic information to model global land-use dynamics throughout the 21st century. MAgPIE operates at two spatial scales that include a coarse scale of twelve model regions with similar socioeconomic characteristics and a subregional level that combines biophysical information, such as suitability for agricultural production (crop yield potential, irrigation water requirements, travel time to urban markets etc.), to determine the cost-effectiveness of different land-use activities. Both these resolutions, however, are typically too coarse to assess how global land-use dynamics could drive landscape-scale changes and associated changes in biodiversity and ecosystems service supply, which is a key sustainability challenge.

The Spatial Economic Allocation Landscape Simulator (SEALS) model can therefore be applied to spatially allocate projected land-use dynamics to spatial scale that is relevant for assessing landscape and biodiversity change. SEALS uses empirically calibrated adjacency relationships, physical suitability and conversion eligibility information at a spatial resolution of 300 x 300 Meter to allocate land use activities across space.

In the following this tutorial describes how both the MAgPIE and SEALS models can be linked using some of the functionalities provided by MAgPIE and SEALS to processes model outputs (MAgPIE) or inputs (SEALS) respectively. This tutorial therefore assumes some familiarity with the MAgPIE model - in particular how to set up model runs. Basic tutorials on how to run the MAgPIE model can be found here <https://magpiemodel.github.io/tutorials/>. A full documentation of the MAgPIE model (release version 4.7.2) can be accessed under <https://rse.pik-potsdam.de/doc/magpie/4.7.2/> (substitute the version number 4.7.2 with any newer release version number to get the latest documentation).

11.2 Create MAgPIE outputs for SEALS by using output scripts

The MAgPIE model provides several output scripts that can be executed after a model run and are stored under `scripts/output` in the MAgPIE folder. Two scripts are particularly

relevant for providing MAgPIE outputs to SEALS: the `extra/disaggregation.R` script and the `extra/reportMAgPIE2SEALS.R` script. The `extra/disaggregation.R` script disaggregates coarse-resolution land-use projections to a spatial resolution of 0.5 degrees and is run by default after each model run. The `extra/reportMAgPIE2SEALS.R` script, on the other hand, generates a NetCDF file that can readily be used by SEALS but is *not* run by default.

The output scripts can be selected and executed via the command line. Therefore, on the command prompt, navigate to the MAgPIE model folder and use the following command:

Rscript output.R

On the command line you are now asked for which of the model runs you would like to do the postprocessing. You can choose **all**, select a specific run, or run the postprocessing for a selection of runs by choosing **search by the pattern**.

```
CLUSTER2015 - vjeetze@login01:/p/projects/magpie/users/vjeetze/magpie/other/f_magpie-seals_tutorial> Rscript output.R
Global .Rprofile loaded! (R version 4.1.2 (2021-11-01))
Hi Pat,

Your .Rprofile was loaded!

Checking for updates... Update check done.
Checking package version requirements... Requirements check done.

Choose runs:
1: all
2: weeklyTests_SSP2EU-NDC/
3: weeklyTests_SSP2EU-PkBudg650/
4: weeklyTests_SSP2EU-Ref/
5: Search by the pattern.
Number:
```

After this you can select the output script that you want to execute. Choose **extra**.

Main selection of MAgPIE output scripts

```
-----
-> Scripts in this selection are actively managed and work out of <-
-> the box. Please provide a yml header to your script specifying <-
-> "description" and "comparison script", otherwise it may not be <-
-> processed correctly. <-
-----
```

1:	output check		check output for known problems
2:	rds report		extract report in rds and mif format from run
3:	merge report		Merges report.mif files from several runs into a single mif file
4:	rds report iso		extract report in rds format from run
5:	validation		creates a validation pdf file for a run (long version - single crop types)
6:	validation short		creates a validation pdf file for a run (short version - aggregated crop types)
7:	comparison validation		Creates an validation pdf out of several runs
8:	runBlackmagicc		Append MAGICC7 warming pathways to a MAgPIE run's report.mif
9:	validation cell		Comparison at cellular level of land and crop types with LUH2 and MAPSPAM

Alternatively, choose a output script from another selection:

10:	extra		Additional MAgPIE output scripts
11:	projects		Project-specific MAgPIE output scripts
12:	deprecated		Deprecated scripts

Choose a output script or folder: 10

Next select reportMAgPIE2SEALS.

Additional MAgPIE output scripts

```

-----
->     Scripts in this selection might require some manual      <-
-> adjustments before they work with the given MAgPIE version as <-
-> they are not necessarily updated along with changes in the   <-
->     model code. Please provide a yml header to your script   <-
-> specifying "description" and "comparison script", otherwise it <-
->         may not be processed correctly.                       <-
-----

1:          aff area | extracts afforestation area from
                   | multiple runs and creates a PDF
                   | (useful template for other
                   | variables)
2:      disaggregation LUH2 | Downscale MAgPIE results to 0.25
                   | degree resolution in LUH2 format for
                   | ISIMIP 3b
3:          disaggregation | Interpolates land pools to 0.5
                   | degree resolution
4:          emulator       |
5:      force runstatistics | Forces runstatistics submission in
                   | central repo
6:      ForestChangeCluster | Compares Forest Area Change at
                   | Cluster Level
7:          highres        | starts a run with higher resolution
                   | in parallel mode (each region is
                   | solved individually) using trade
                   | patterns from an existing run
8:          land cluster   | Write land use information on
                   | cluster level to land_cluster.gpkg
                   | (GeoPackage)
9:          modelstat      | checks the modelstat of several runs
10: reportMAgPIE2REMIND    | Write only those variables into a
                   | report that are relevant for the
                   | REMIND coupling
11:      reportMAgPIE2SEALS | Modifies gridded MAgPIE land use
                   | patterns so that they can be read in
                   | by the Spatial Economic Allocation
                   | Landscape Simulator (SEALS)
12:          resubmit      | re-submits runs to different queues
                   | (e.g. priority)
13:          runtime       | Checks the runtime of several runs
14:      timestep duration | Makes a png file containing solve
                   | time for each step
15: validation existing report | Creates a validation pdf out of an
                   | existing mif file, speeding up the
                   | process.

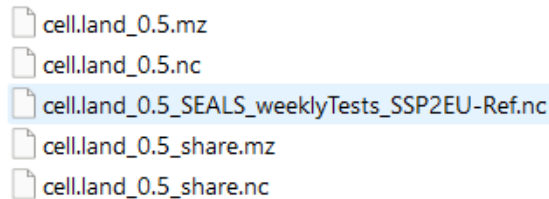
Choose a script: 11

```

Finally, choose the run submission type, e.g. `Direct execution`.

```
Choose submission type:
1: SLURM standby
2: SLURM standby maxMem
3: SLURM priority
4: SLURM priority maxMem
5: Direct execution
6: Background execution
7: Debug mode
Number: 5
```

All outputs are written to a subfolder in the `output` folder within the MAgPIE directory. This subfolder (or run folder) is created automatically at the beginning of a model run and its name is a combination of the run title and date (For further information see <https://mag-piemodel.github.io/tutorials/t06-changingconfig>). Before you run `reportMAgPIE2SEALS`, please make sure that the script `extra/disaggregation.R` was executed without error. If the script `extra/disaggregation.R` was completed successfully, the run folder should contain the file `cell.land_0.5_share.mz`. Once the script `extra/reportMAgPIE2SEALS.R` has finished the run folder should contain a file named `cell.land_0.5_SEALS_<run title>.nc`. This file contains spatially-explicit information on the share of total land per grid cell used for different land-use activities (cropland, pasture, forest, urban etc.) projected until 2050, which can now be used as an input for SEALS.



```
cell.land_0.5.mz
cell.land_0.5.nc
cell.land_0.5_SEALS_weeklyTests_SSP2EU-Ref.nc
cell.land_0.5_share.mz
cell.land_0.5_share.nc
```

11.2.1 Change your start script to generate MAgPIE outputs for SEALS

If you are familiar with setting up a start script for MAgPIE runs (see <https://mag-piemodel.github.io/tutorials/t07-startscript>), an alternative way of generating MAgPIE outputs for SEALS is by adding `"extra/reportMAgPIE2SEALS"` to the setting

```
cfg$output <- c("output_check", "extra/disaggregation", "rds_report")
```

so that it reads

```
cfg$output <- c("output_check", "extra/disaggregation", "rds_report", "extra/reportMAgPIE2SEALS")
```

As the order of the vector determines the order of execution, please make sure that "extra/reportMagPIE2SEALS" is called after "extra/disaggregation".

11.3 Setting up SEALS to downscale MAgPIE outputs

After generating the output `cell.land_0.5_SEALS_<run title>.nc`, the following describes how this information can be used as input for the SEALS model.

First, we need to relate the land use types of the MAgPIE model to the land cover types used by the SEALS model. While MAgPIE separates seven different main land types (cropland, grassland, primary forest, secondary forest, forestry, non-forest vegetation & urban), the SEALS model only allocates changes for 5 main land types (urban, cropland, grassland, forest & other natural vegetation). Below a mapping between the different land types is shown.

```
src_id,dst_id,src_label,dst_label,src_description,dst_description
6,1,urban,urban,urban land,
1,2,crop,cropland,cropland,
2,3,past,grassland,grassland,
3,4,primforest,forest,primary forest,
4,4,secdforest,forest,secondary forest,
5,4,forestry,forest,forestry,
7,5,other,othernat,non-forest vegetation,
```

Copy and paste this mapping into a CSV file and save it, e.g. under `base_data/seals/default_inputs/magpie_s`. The folder `base_data/seals/default_inputs` also contains other mappings such as a mapping between LUH2v2 land types and SEALS. This mapping can now be used to set up the **scenarios CSV**. The scenarios CSV provides SEALS with all necessary information on the input data (or MAgPIE outputs), including the path to the MAgPIE outputs. It also allows the user to set up multiple SEALS runs (e.g. in the case of different scenario runs) in one go. The file is named per project — the one shipped with SEALS is `standard_scenarios.csv` — and your run file points at it with `p.scenario_definitions_filename`. One way to better understand it is to run SEALS on the test data without further modification, which generates a default in the project folder that you can then tailor to your needs. The columns are documented in [scenarios_format.md](#).

The following shows how an example `scenario_definitions_tutorial.csv` is created that provides relevant information for a set of different MAgPIE test runs. Assume we have created land use projections for three different test scenarios: a reference ‘business-as-usual’ scenario (`weeklyTests_SSP2EU-Ref`), a low ambition mitigation scenario, in which current nationally determined contributions (NDCs) under the Paris agreement for the land sector are fulfilled (`weeklyTests_SSP2EU-NDC`) and a highly ambitious mitigation scenario in line with the 1.5 degree target (`weeklyTests_SSP2EU-PkBudg650`). For each of the scenarios, we have created

gridded land-use projections in a format that can be read by SEALS using the output script `extra/reportMagPIE2SEALS.R` (see above).

Here, the `scenario_definitions_tutorial.csv` contains a line for each scenario specified under `scenario_label`. Note that the `Baseline` always needs to be defined as well. In the `years` column you can specify the years for which SEALS should do a downscaling. Multiple years can be separated by a space.

	column 1	column 2	column 3	column 4	column 5	column 6	column 7	column 8
1	scenario_label	scenario_type	aoi	exogenous_label	climate_label	model_label	counterfactual_label	years
2	Baseline	baseline	global	baseline		magpie		2020
3	weeklyTests_SSP2EU-Ref	bau	global	ssp2	rcp26	magpie	weeklyTests_SSP2EU-Ref	2030 2050
4	weeklyTests_SSP2EU-NDC	policy	global	ssp2	rcp26	magpie	weeklyTests_SSP2EU-NDC	2030 2050
5	weeklyTests_SSP2EU-PkBudg650	policy	global	ssp2	rcp26	magpie	weeklyTests_SSP2EU-PkBudg650	2030 2050

Set the path to the gridded MAGPIE outputs in the column `coarse_projections_input_path`

	column 1	column 14	column 15
1	scenario_label	coarse_projections_input_path	lulc_src_label
2	Baseline	C:/Users/vjeetze/PIK/MAGPIE/other/magpie_seals_tutorial/cell.land_0.5_SEALS_weeklyTests_SSP2EU-Ref.nc	esa
3	weeklyTests_SSP2EU-Ref	C:/Users/vjeetze/PIK/MAGPIE/other/magpie_seals_tutorial/cell.land_0.5_SEALS_weeklyTests_SSP2EU-Ref.nc	esa
4	weeklyTests_SSP2EU-NDC	C:/Users/vjeetze/PIK/MAGPIE/other/magpie_seals_tutorial/cell.land_0.5_SEALS_weeklyTests_SSP2EU-NDC.nc	esa
5	weeklyTests_SSP2EU-PkBudg650	C:/Users/vjeetze/PIK/MAGPIE/other/magpie_seals_tutorial/cell.land_0.5_SEALS_weeklyTests_SSP2EU-PkBudg650.nc	esa

The `lulc_correspondence_path` gives the path to a mapping between ESA CCI land cover classes and the aggregated land cover types used in SEALS.

	column 1	column 17	column 18	column 19
1	scenario_label	lulc_correspondence_path	coarse_src_label	coarse_simplification_label
2	Baseline	seals/default_inputs/esa_seals7_correspondence_mosaic-natural.csv	magpie	seals7
3	weeklyTests_SSP2EU-Ref	seals/default_inputs/esa_seals7_correspondence_mosaic-natural.csv	magpie	seals7
4	weeklyTests_SSP2EU-NDC	seals/default_inputs/esa_seals7_correspondence_mosaic-natural.csv	magpie	seals7
5	weeklyTests_SSP2EU-PkBudg650	seals/default_inputs/esa_seals7_correspondence_mosaic-natural.csv	magpie	seals7

The default mapping in SEALS is stored under `base_data\seals\default_inputs\esa_seals7_correspondence`. For MAGPIE-SEALS couplings, however, it is recommended to use a mapping that classifies “crop_natural_mosaic” as “othernat” and “natural_crop_mosaic” as “forest” to better align the total cropland extent of the ESA map with the cropland initialisation in MAGPIE. That there is an overestimation of the cropland extent in the ESA CCI data is a well-know issue. Thus, the lines in `esa_seals7_correspondence.csv`

```
6,,30,2,crop_natural_mosaic,cropland
7,,40,2,natural_crop_mosaic,cropland
```

are changed to

```
6,,30,2,crop_natural_mosaic,othernat
7,,40,2,natural_crop_mosaic,forest
```

in `esa_seals7_correspondence_mosaic-natural.csv`.

Next, we set the path to the mapping between MAgPIE land use types to SEALS land cover types, as defined above

	column 1	column 20	column 21	column 22
1	scenario_label	coarse_correspondence_path	lc_class_varname	dimensions
2	Baseline	seals/default_inputs/maggpie_seals7_correspondence.csv	all_variables	time
3	weeklyTests_SSP2EU-Ref	seals/default_inputs/maggpie_seals7_correspondence.csv	all_variables	time
4	weeklyTests_SSP2EU-NDC	seals/default_inputs/maggpie_seals7_correspondence.csv	all_variables	time
5	weeklyTests_SSP2EU-PkBudg650	seals/default_inputs/maggpie_seals7_correspondence.csv	all_variables	time

We also specify a set of regressors used during downscaling. Tables of trained model coefficients used by SEALS can be found under `base_data/seals/default_inputs`. Here we can also align some model constraints between MAgPIE and SEALS. If, for example, compliance with currently implemented land policies such as protected areas is assumed in MAgPIE, these constraints also need to be reflected during the downscaling. We here therefore add an additional multiplicative regressor that precludes expansion of agricultural land (cropland and grassland) in legally protected areas as reported by the WDPA database, which is in line with the assumption in MAgPIE.

	column 1	column 2	column 3	column 4	column 5	column 6	column 7	column 8
1	spatial_regressor_name	data_location	type	urban	cropland	grassland	forest	othernat
11	grassland_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_grassland.tif	additive	0.005555556	0.018888889	0	0.041666667	-0.026111111
12	forest_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_forest.tif	additive	-0.019444444	-0.016666667	-0.002666667	0	0.033444444
13	othernat_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_othernat.tif	additive	0.01	0.144444444	0.060111111	0.02	0
14	water_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_water.tif	additive	0	0	0	0	0
15	other_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_other.tif	additive	-1.119444444	0.001666667	0.126666667	0.061111111	-0.023333333
16	urban_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_url	additive	1.713888889	-1122.233444	-11.130555556	0.041666667	-1122.241667
17	cropland_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_cr	additive	0.105555556	0.333444444	0.022222222	0	-11.24444444
18	grassland_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_gr	additive	0.054444444	0.005444444	0.38	0.018222222	0.085555556
19	forest_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_fo	additive	-0.122222222	1111.065556	-0.011	1.555	-0.022233333
20	othernat_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_otl	additive	0.010888889	0	0.019444444	-0.122222222	0.466666667
21	water_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_wa	additive	0.036555556	-112.2027778	-112.24	-1.105555556	-1133.322333
22	other_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_otl	additive	-0.127777778	-112.2555556	-112.144444	-1111.133333	0.005555556
23	urban_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_url	additive	-0.072222222	-11.52222222	-11.2638889	-0.093333333	-0.087777778
24	cropland_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_cr	additive	0.068888889	0.162333333	-0.016666667	0.122222222	0.065555556
25	grassland_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_gr	additive	0.100222222	-0.025	0.431111111	-0.026677778	-0.041666667
26	forest_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_fo	additive	0.133222222	0.367777778	0.076333333	0.68888	0.113333333
27	othernat_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_otl	additive	0	-0.073111111	0.024444444	-0.005555556	0.152777778
28	water_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_wa	additive	0.091666667	0.005	-1111.105444	-1.092777778	-0.002222222
29	other_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_otl	additive	0.045555556	0.15	-1111.077778	-0.008333333	-110.89
30	soil_organic_content_1m_30s	seals/static_regressors/soil_organic_content.tif	additive	0.027777778	-0.15	110.977778	-111.14	-0.027777778
31	bio_12	seals/static_regressors/precip_mm.tif	additive	11.11944444	-0.994444444	1.14	-1.075	11.00444444
32	alt	seals/static_regressors/alt_m.tif	additive	-0.104444444	0.085	-0.024888889	-0.037788889	0.01
33	bio_1	seals/static_regressors/temperature_c.tif	additive	-0.022111111	0.044444444	-0.011111111	-0.01	-0.001111111
34	minutes_to_market_30s	seals/static_regressors/travel_time_to_market_mins.tif	additive	0.016122222	0.21	0.005555556	10.555	-0.034333333
35	pop_30s	seals/static_regressors/pop.tif	additive	0	0	0	0	0
36	CEC_1m_30s	seals/static_regressors/soil_cec.tif	additive	0	-0.016666667	0	111.0722222	0
37	clay_percent_1m_30s	seals/static_regressors/clay_percent.tif	additive	-0.051111111	0.02	-0.186111111	-0.046222222	1.084333333
38	ph_1m_30s	seals/static_regressors/ph.tif	additive	0	0.1	0	0	0
39	sand_percent_1m_30s	seals/static_regressors/sand_percent.tif	additive	0.034444444	0.018333333	-0.037777778	-0.048888889	-0.001111111
40	silt_percent_1m_30s	seals/static_regressors/silt_percent.tif	additive	-0.012777778	-0.165	0	-0.059011111	-0.146111111
41	wdpa	seals/static_regressors/maggpie_WDPA.tif	multiplicative	0	0	0	1	1
42	wetlands_binary	seals/static_regressors/wetlands_binary.tif	additive	0	0	0	0	0
43	carbon_above_ground_mg_per_ha_global	seals/static_regressors/carbon_above_ground_mg_per_ha_global.tif	additive	0	0	0	0	0

Similarly, the same functionality can also be used to specify policy scenarios, such as the expansion of protected areas in line with the 30x30 goal of the Global Biodiversity Framework. The `seals/static_regressors/maggpie_30by30.tif` contains both currently protected areas from the WDPA database and areas that might be protected under a global 30x30 protection scenario.

	column 1	column 2	column 3	column 4	column 5	column 6	column 7	column 8
1	spatial_regressor_name	data_location	type	urban	cropland	grassland	forest	othernat
11	grassland_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_grassland.tif	additive	0.005555556	0.018888889	0	0.041666667	-0.026111111
12	forest_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_forest.tif	additive	-0.019444444	-0.016666667	-0.002666667	0	0.033444444
13	othernat_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_othernat.tif	additive	0.01	0.144444444	0.060111111	0.02	0
14	water_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_water.tif	additive	0	0	0	0	0
15	other_binary	lulc/esa/seals7/binaries/2014/binary_esa_seals7_2014_other.tif	additive	-1.119444444	0.001666667	0.126666667	0.061111111	-0.023333333
16	urban_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_urb.tif	additive	1.713888889	-112.233444	-11.13055556	0.041666667	-112.241667
17	cropland_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_cro.tif	additive	0.105555556	0.333444444	0.022222222	0	-11.24444444
18	grassland_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_gra.tif	additive	0.054444444	0.005444444	0.38	0.018222222	0.085555556
19	forest_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_for.tif	additive	-0.122222222	1111.065556	-0.011	1.555	-0.022233333
20	othernat_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_oth.tif	additive	0.010888889	0	0.019444444	-0.122222222	0.466666667
21	water_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_wa.tif	additive	0.036555556	-112.2027778	-112.24	-1.105555556	-1133.322333
22	other_gaussian_1	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_oth.tif	additive	-0.127777778	-112.2555556	-111.2.144444	-111.1.133333	0.005555556
23	urban_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_urb.tif	additive	-0.072222222	-11.52222222	-11.2638889	-0.093333333	-0.087777778
24	cropland_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_cro.tif	additive	0.068888889	0.162333333	-0.016666667	0.122222222	0.065555556
25	grassland_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_gra.tif	additive	0.100222222	-0.025	0.431111111	-0.026677778	-0.041666667
26	forest_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_for.tif	additive	0.133222222	0.367777778	0.076333333	0.68888	0.113333333
27	othernat_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_oth.tif	additive	0	-0.073111111	0.024444444	-0.005555556	0.152777778
28	water_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_wa.tif	additive	0.091666667	0.005	-111.105444	-1.092777778	-0.002222222
29	other_gaussian_5	lulc/esa/seals7/convolutions/2014/convolution_esa_seals7_2014_oth.tif	additive	0.045555556	0.15	-111.077778	-0.008333333	-10.89
30	soil_organic_content_1m_30s	seals/static_regressors/soil_organic_content.tif	additive	0.027777778	-0.15	110.977778	-111.14	-0.027777778
31	seals_12	seals/static_regressors/precip_mm.tif	additive	11.11944444	-0.994444444	1.14	1.075	11.00444444
32	alt	seals/static_regressors/alt_m.tif	additive	-0.104444444	0.085	-0.024888889	-0.037788889	0.01
33	bio_1	seals/static_regressors/temperature_c.tif	additive	-0.022111111	0.044444444	-0.011111111	-0.01	-0.001111111
34	minutes_to_market_30s	seals/static_regressors/travel_time_to_market_mins.tif	additive	0.016122222	0.21	0.005555556	10.555	-0.034333333
35	pop_30s	seals/static_regressors/pop.tif	additive	0	0	0	0	0
36	CEC_1m_30s	seals/static_regressors/soil_cec.tif	additive	0	-0.016666667	0	111.0722222	0
37	clay_percent_1m_30s	seals/static_regressors/clay_percent.tif	additive	-0.051111111	0.02	-0.186111111	-0.046222222	1.084333333
38	ph_1m_30s	seals/static_regressors/ph.tif	additive	0	0.1	0	0	0
39	sand_percent_1m_30s	seals/static_regressors/sand_percent.tif	additive	0.034444444	0.018333333	-0.037777778	-0.048888889	-0.001111111
40	silt_percent_1m_30s	seals/static_regressors/silt_percent.tif	additive	-0.012777778	-0.165	0	-0.059011111	-0.146111111
41	30by30	seals/static_regressors/magpie_30by30.tif	multiplicative	0	0	0	1	1
42	wetlands_binary	seals/static_regressors/wetlands_binary.tif	additive	0	0	0	0	0
43	carbon_above_ground_mg_per_ha_global	seals/static_regressors/carbon_above_ground_mg_per_ha_global.tif	additive	0	0	0	0	0

Finally, the path to the regressor used during the downscaling must also be added to the scenarios CSV, while the base land-cover data `base_year_lulc_path` must refer to the same year as defined in the **Baseline** case.

	column 1	column 20	column 21	column 22	column 23	column 24
1	scenario_label	espondence_path	lc_class_varname	dimensions	calibration_parameters_source	base_year_lulc_path
2	Baseline	lt_inputs/magpie_seals7_correspondence.csv	all_variables	time	seals/default_inputs/magpie_global_coefficients_wdpa_base.csv	lulc/esa/lulc_esa_2020.tif
3	weeklyTests_SSP2EU-Ref	lt_inputs/magpie_seals7_correspondence.csv	all_variables	time	seals/default_inputs/magpie_global_coefficients_wdpa_base.csv	lulc/esa/lulc_esa_2020.tif
4	weeklyTests_SSP2EU-NDC	lt_inputs/magpie_seals7_correspondence.csv	all_variables	time	seals/default_inputs/magpie_global_coefficients_wdpa_base.csv	lulc/esa/lulc_esa_2020.tif
5	weeklyTests_SSP2EU-PkBudg650	lt_inputs/magpie_seals7_correspondence.csv	all_variables	time	seals/default_inputs/magpie_global_coefficients_wdpa_base.csv	lulc/esa/lulc_esa_2020.tif

Save `scenario_definitions_tutorial.csv` in the `input_template/` directory beside your run file — it is tracked in git, and ProjectFlow copies it into the project's `input/` on first run. Then name it in the run file's `__main__` block with `p.scenario_definitions_filename = 'scenario_definitions_tutorial.csv'` and you are good to go.

```

71 sys.path.insert(0, custom_seals_path)
72
73 # SEALS will run based on the scenarios defined in a scenario_definitions.csv
74 # If you have not run SEALS before, SEALS will generate it in your project's input_dir.
75 # A useful way to get started is to run SEALS on the test data without modification
76 # and then edit the scenario_definitions.csv to your project needs.
77 # Some of the other test files use different scenario definition csvs
78 # to illustrate the technique. If you point to one of these
79 # (or any one CSV that already exists), SEALS will not generate a new one.
80 # The available example files in the default_inputs include:
81 # - test_three_scenario_defininitions.csv
82 # - test_scenario_defininitions_multi_coeffs.csvs
83
84 p.scenario_definitions_path = os.path.join(p.input_dir, 'scenario_definitions_tutorial.csv')
85
86 # Set defaults and generate the scenario definitions csv if it doesn't exist

```

Part II

Worked examples

12 Worked example: Kenya DEMETRA-SEALS-InVEST

Workflow Walkthrough

i Historical report — not a current guide

This documents the DEMETRA-SEALS-InVEST workflow **as it stood in July 2026**. It is kept as a record of that work and is not maintained against the current API. The run file it describes, `run_ken_demetra.py`, and the `ecosystem_services.py` module it used have since been retired to `seals/seals/old_run_files/` and are no longer supported. Run files have also moved to the `run_project(p)` convention since this was written, so the code shown here will not match what is in the repo. For a current walkthrough see [SEALS Quickstart](#) and [First SEALS run walkthrough](#).

12.1 Overview

This walkthrough guides you through installing, configuring, and running the DEMETRA-SEALS-InVEST workflow. SEALS (Spatial Economic Allocation Landscape Simulator) integrates with InVEST (Integrated Valuation of Ecosystem Services and Tradeoffs) to model land-use change scenarios and their ecosystem service impacts.

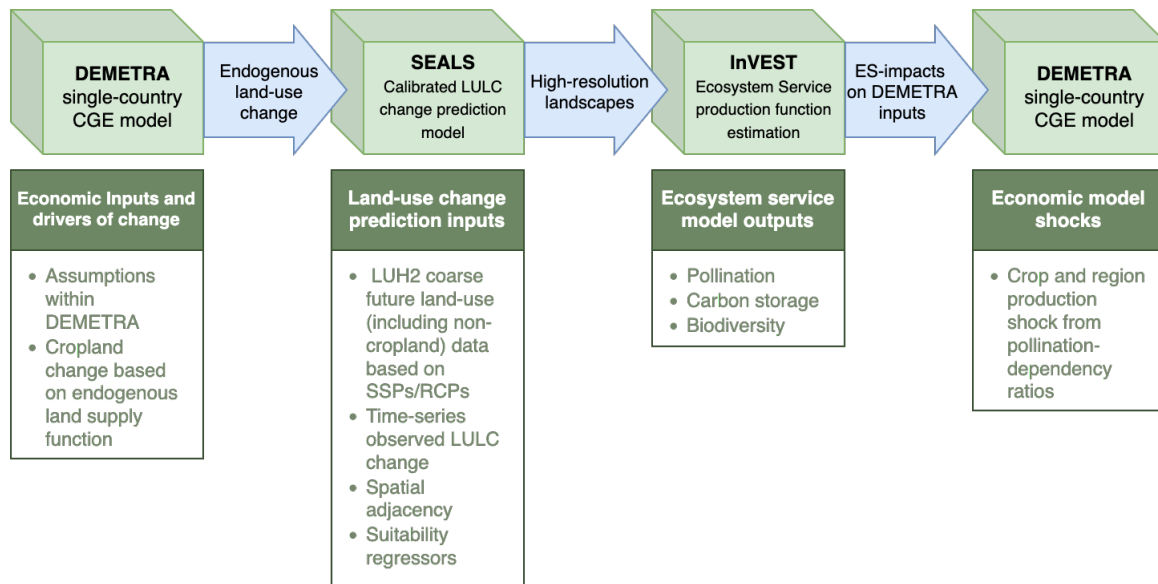


Figure 12.1: Workflow Diagram

12.2 Installation

For complete details, see Justin Johnson's [installation guide](#).

i Prerequisites

Before beginning, ensure you have:

- **Git** - [Download here](#)
- **Miniforge3** (Python distribution) - [Download here](#)

i Basic Hazelbean Installation

Hazelbean is the core dependency for running SEALS. Install it with these commands:

```
# Initialize conda (may not be needed depending on your shell configuration)
conda init

# Create a new environment (replace 'seals_env' with your preferred name)
conda create -n seals_env

# Activate the environment
conda activate seals_env

# Install Hazelbean
mamba install hazelbean
```

Note: The `mamba install hazelbean` command may take 5-10 minutes to complete.

12.2.1 SEALS Repository Installation

i Step 1: Install C/C++ Compiler

Windows Users - Choose One Option:

- **Option 1:** Download from [Visual Studio Build Tools](#) and select “Download Build Tools”
- **Option 2:** Use winget command:

```
winget install Microsoft.VisualStudio.2022.BuildTools --force --override "--passive --
```

- **Option 3:** Run the `install.bat` file from the Earth Economy Devstack repository root (executes the winget command above)

Mac/Linux Users:

Most users will already have Xcode installed. If not:

```
xcode-select --install
```

This installs Xcode Command Line Tools including gcc and clang compilers.

i Step 2: Clone SEALS Repository

Clone the repository to your local machine:

```
git clone https://github.com/jandrewjohnson/seals
```

Recommended location: C:/Users/<username>/Files/seals/seals (Windows) or /Users/<username>/Files/seals/seals (Mac/Linux)

i Step 3: Install as Editable Package

```
# Activate your conda environment
conda activate seals_env

# Navigate to the cloned repository
cd C:/Users/<username>/Files/seals/seals

# Install as editable package
pip install -e . --no-deps
```

12.3 Data Setup

12.3.1 File Organization Structure

SEALS uses a specific directory structure to locate data automatically:

12.3.1.1 Base Data (Global Shared Data)

SEALS automatically downloads base data on first run to:

```
<home>/Files/base_data/seals/
<home>/Files/base_data/lulc/
```

12.3.1.2 Additional InVEST data

```
<home>/Files/base_data/global_invest/
<home>/Files/base_data/crops/
```

12.3.1.3 Project Data (Country/Region-Specific)

12.3.1.4 Additional country data

<home>/Files/seals/projects/run_ken_demetra/

This includes:

- Region border files (shapefiles and raster GeoTIFFs)
 - DEMETRA shock files (CSV)
 - Downscaling coefficient files (CSV and global rasters)
 - Scenario configuration files
-

12.4 Testing SEALS

12.4.1 Standard Test Run

This basic test verifies your installation is working correctly.

Steps:

1. Locate `run_seals_standard.py` in `seals/seals/`
2. Ensure your conda environment is activated:

```
conda activate seals_env
```

3. Run the script:

```
python run_seals_standard.py
```

4. Check for output in:

<home>/Files/seals/projects/standard_<timestamp>/intermediate/visualization/lulc_pngs/

Expected Output: You should see land-use/land-cover (LULC) maps like:

Lulc Ssp2 Rcp45 Luh2-Message Bau 2050

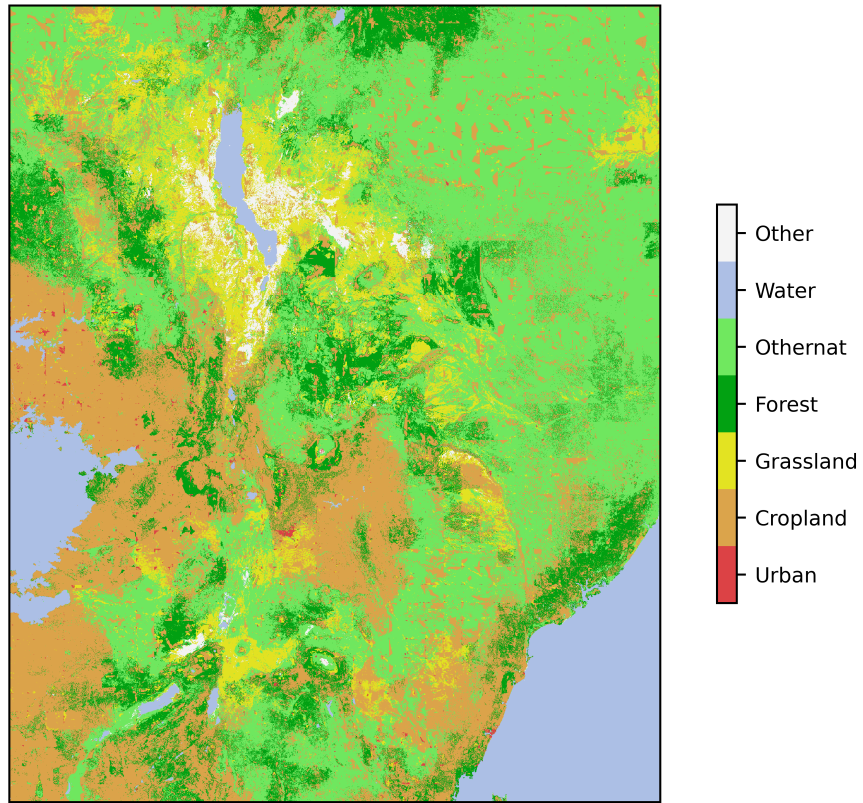


Figure 12.2: Example Output

12.4.2 Common Issues

Problem: `ValueError: Could not open...` error on first run

Solution: The `base_data` download likely didn't complete. Run the script again now that `base_data` is fully downloaded.

12.5 Kenya DEMETRA Scenario

This scenario demonstrates land-use change modeling for Kenya using DEMETRA economic shocks.

12.5.1 Key Input Files

Scripts:

- Main execution script: `run_ken_demetra.py`

Data Files:

File	File Type	Location
Scenario Configuration	<code>ken_scenarios_full.csv</code>	<code>projects/run_ken_demetra/</code>
Default Downscaling	CSV files + Global rasters	<code>base_data/seals/static_regressors</code>
Coefficients		<code>base_data/seals/default_inputs</code>
DEMETRA Shocks	CSV files	<code>projects/run_ken_demetra/inputs/demetra</code>
Regional Borders	Shapefiles + Raster GeoTIFFs	<code>projects/run_ken_demetra/inputs/borders</code>
Project Downscaling	CSV files + Global rasters	<code>projects/run_ken_demetra/inputs/static_</code>
Coefficients		<code>projects/run_ken_demetra/inputs/coefficient</code>

Note: The structure is flexible, but ensure that the paths defined in the scenario CSV and coefficients CSV are properly defined.

12.5.2 Running the Kenya Scenario

```
conda activate seals_env
cd seals/seals/
python run_ken_demetra.py
```

12.6 Understanding the Task Tree

When you run `run_ken_demetra.py`, SEALS creates a `ProjectFlow` object that executes a series of tasks in a hierarchical tree structure. Each task creates outputs in the `intermediate/` directory.

12.6.1 Task Hierarchy

For reference, here's the complete task tree as shown in the console output:

```
execute
  project_aoi
  fine_processed_inputs
    generated_kernels
    lulc_clip
    lulc_simplifications
    lulc_binaries
    lulc_convolution
  coarse_change
    coarse_extraction
    coarse_simplified_proportion
    coarse_simplified_ha
    coarse_simplified_ha_difference_from_previous_year
  regional_change
  allocations
    allocation_zones
    allocation
  stitched_lulc_simplified_scenarios
  visualization
    lulc_pngs
  reproject_lulc_rasters_to_equal_area
  run_invest_carbon
  run_invest_pollination
  calculate_crop_value_and_shock
  calculate_biodiversity_index
  summarize_and_visualize_multi_vector
```

Each task is explained in detail below.

12.7 Task-by-Task Walkthrough

12.7.1 1. Project AOI (Area of Interest)

Significance: Defines the geographic boundary for the analysis and creates resolution pyramids for efficient processing at multiple scales.

What it does:

1. Loads the area of interest boundary for Kenya from the input shapefile
2. Creates a vector geopackage (**.gpkg**) file for the AOI
3. Generates raster “pyramids” at different resolutions:
 - Fine resolution: Matches the base LULC data (typically 300m)
 - Coarse resolution: Matches the coarse projection data (typically 0.25°)

Input files:

- Region shapefile from `projects/run_ken_demetra/inputs/borders`

Output files:

```
intermediate/project_aoi/  
  aoi_KEN.gpkg          # Vector boundary  
  pyramids/  
    aoi_ha_per_cell_coarse.tif    # Coarse resolution (hectares per cell)  
    aoi_ha_per_cell_fine.tif      # Fine resolution (hectares per cell)
```

Significance:

- The AOI defines the study region
 - Can use default global country shapefile or project-specific
- Pyramids allow multi-scale analysis
- Fine resolution preserves local detail
- Coarse resolution matches global projections

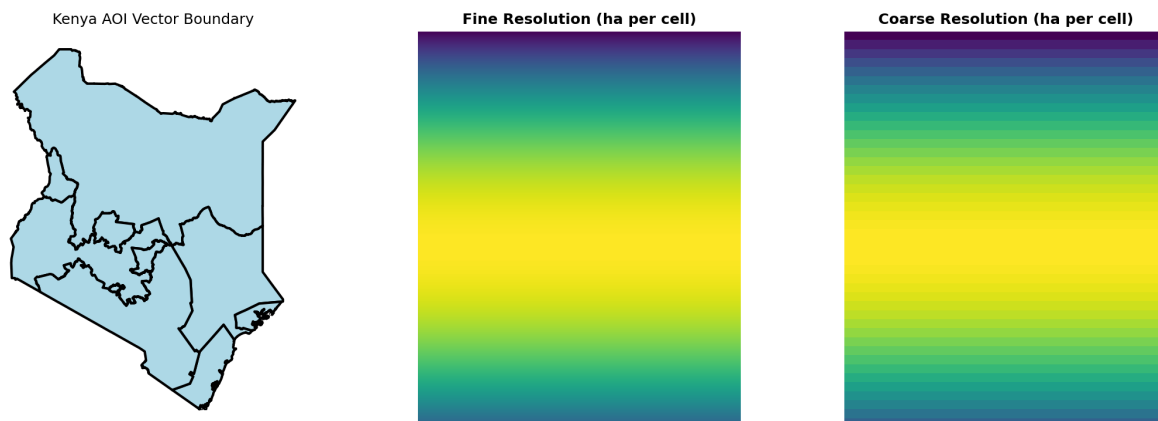


Figure 12.3: Area of Interest for Kenya with resolution pyramids

12.7.2 2. Fine Processed Inputs

Significance: Prepares the baseline land-use/land-cover (LULC) data for allocation by simplifying classifications, creating binary masks, and generating spatial convolutions.

What it does:

1. Generated Kernels: Creates Gaussian kernels for spatial smoothing
2. LULC Clip: Clips global LULC data to the Kenya AOI
3. LULC Simplifications: Reclassifies detailed ESA categories into 7 SEALS classes:
 - Cropland
 - Forest
 - Grassland
 - Urban
 - Water
 - Other natural (othernat)
 - Other
4. LULC Binaries: Creates separate binary masks for each class (1 = present, 0 = absent)
5. LULC Convolutions: Applies Gaussian smoothing to capture neighborhood characteristics

Key Output Structure:

```
intermediate/fine_processed_inputs/
  generated_kernels/
    gaussian_1.tif          # Small neighborhood kernel
    gaussian_5.tif          # Large neighborhood kernel
  lulc/esa/
    lulc_esa_2020.tif       # Original ESA LULC
  seals7/
    lulc_esa_seals7_2020.tif # Simplified 7-class LULC
    binaries/2020/          # Binary masks by class
      binary_esa_seals7_2020_cropland.tif
      binary_esa_seals7_2020_forest.tif
      ...
    convolutions/2020/      # Smoothed neighborhood data
      convolution_esa_seals7_2020_cropland_gaussian_1.tif
      convolution_esa_seals7_2020_cropland_gaussian_5.tif
      ...
```

Significance:

- Simplification reduces complexity while preserving key land-use patterns
- Binary masks enable class-specific analysis
- Convolutions capture spatial context
- These features inform the allocation model's decision-making

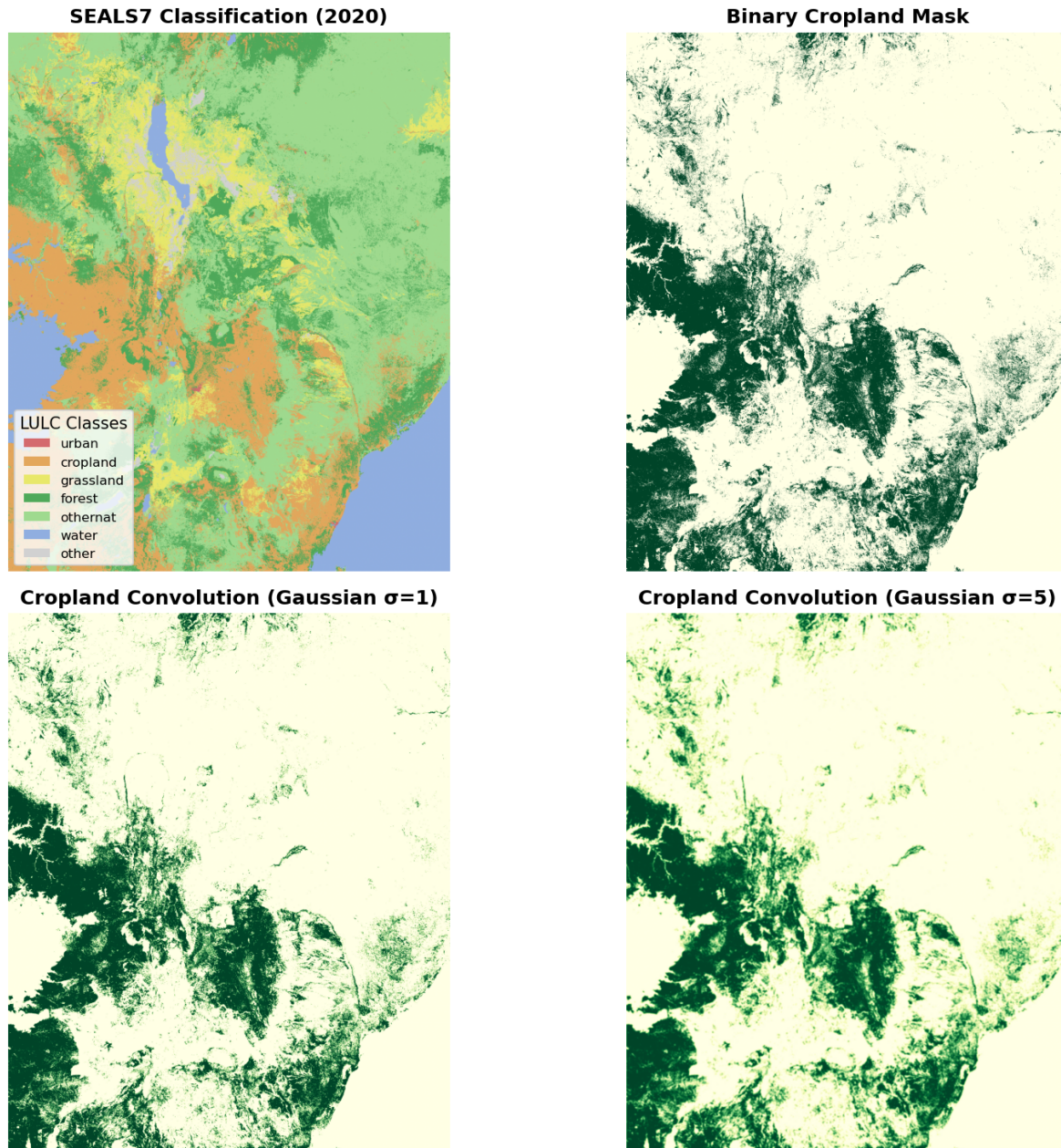


Figure 12.4: Fine processed inputs example: Cropland processing pipeline

12.7.3 3. Coarse Change

Significance: Extracts and processes land-use change projections from global coarse-resolution models (e.g., LUH2) for the Kenya region.

What it does:

1. Coarse Extraction: Clips global projection data to Kenya's bounding box and selected year
2. Coarse Simplified Proportion: Calculates proportions of each SEALS7 class
3. Coarse Simplified Ha: Converts proportions to hectares
4. Ha Difference: Calculates change in hectares between time periods (e.g., 2035 $\rightarrow$ 2050)

Key Output Example:

```
intermediate/coarse_change/coarse_simplified_ha_difference_from_previous_year/  
ssp2/rcp45/luh2-message/bau/2050/  
  cropland_2050_2035_ha_diff_ssp2_rcp45_luh2-message_bau.tif  
  forest_2050_2035_ha_diff_ssp2_rcp45_luh2-message_bau.tif  
  ...
```

Significance:

- Links exogenous global scenarios (SSP2, RCP4.5) to local allocation
- Tracks changes over time (2035 baseline $\rightarrow$ 2050 projection)

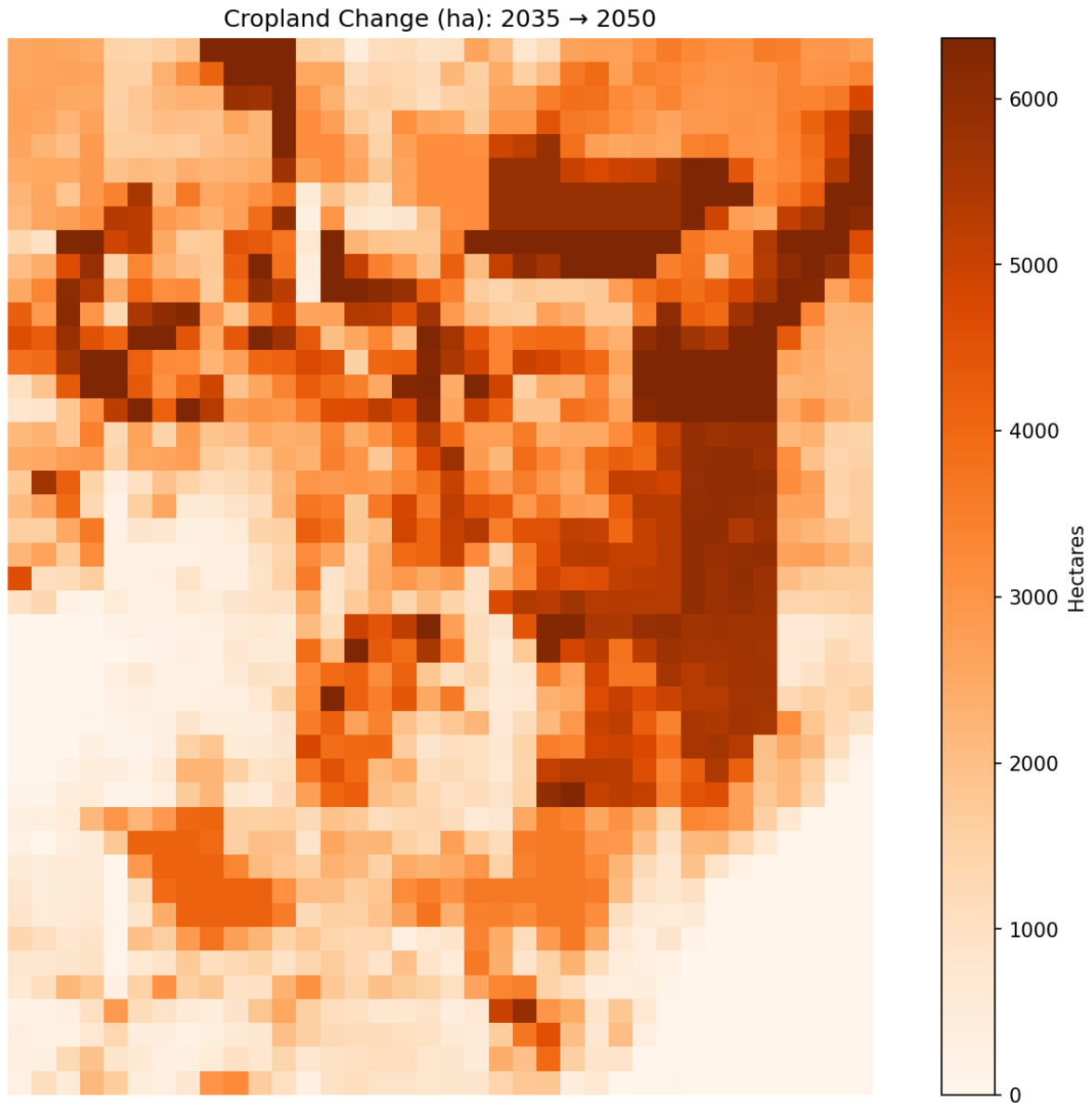


Figure 12.5: Coarse-resolution cropland change (2035-2050) under LP+FP scenario

12.7.4 4. Regional Change

Significance: Adjusts coarse-resolution changes using regional covariates to account for local constraints and opportunities.

What it does:

1. Creates a region ID map dividing AOI into separately shocked zones
2. Applies various algorithms (e.g., proportional, covariate sum shift) to downscale from regional to coarse

Key Outputs:

```
intermediate/regional_change/  
  region_ids.tif                                # Regional zone identifiers  
ssp2/rcp45/luh2-message/LP/2050/  
  cropland_2050_2035_ha_diff_ssp2_rcp45_luh2-message_LP_covariate_sum_shift.tif
```

Significance:

- Accounts for local factors (infrastructure, topography, current land use)
- Bridges the gap between coarse change and regional change

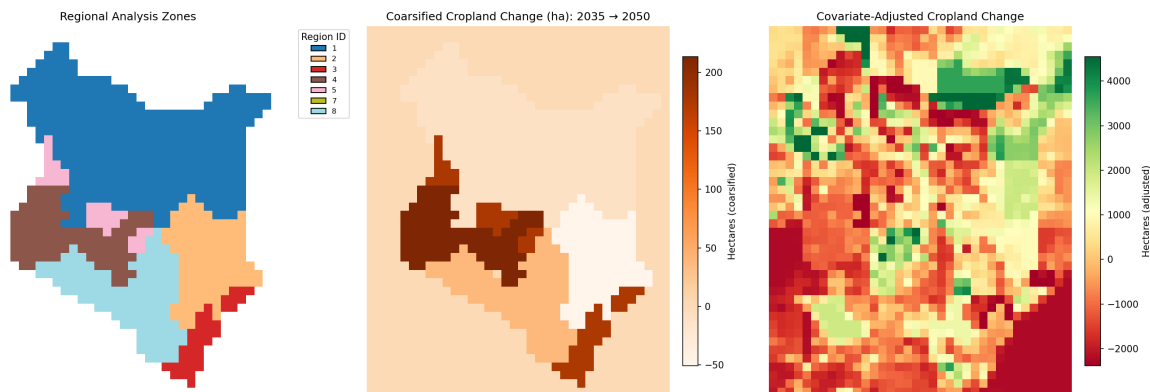


Figure 12.6: Regional + Coarse change for cropland allocation

12.7.5 5. Allocations

Significance: Performs the core SEALS allocation, distributing coarse-resolution changes to fine-resolution pixels based on suitability.

What it does:

1. Divides Kenya into allocation zones (e.g., zone 213_84)
2. For each zone, runs an allocation algorithm that:

- Considers spatial covariates (convolutions, distance, etc.)
- Respects constraints (protected areas, water bodies)
- Tracks cumulative change

3. Produces new LULC maps for each scenario and year

Key Output Structure:

```
intermediate/allocations/ssp2/rcp45/luh2-message/bau/2050/
allocation_zones/213_84/allocation/
    block_ha_per_cell_coarse.tif           # Coarse resolution for this block
    cumulative_change_happened.tif        # Tracking allocated changes
    lulc_esa_seals7_ssp2_rcp45_luh2-message_bau_2050.tif # New LULC
```

Significance:

- This is the core of SEALS, where changes actually get placed on the landscape
- Allocation uses machine learning and optimization to find the “best” locations
- Results are spatially explicit: you can see exactly where cropland expanded

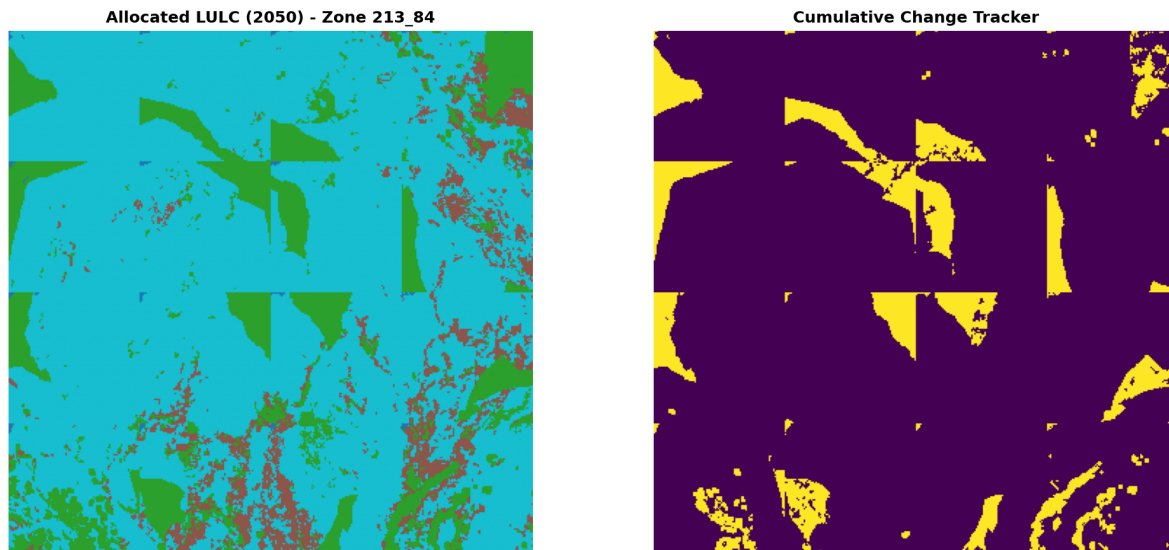


Figure 12.7: Example allocation zone showing LULC projection and cumulative change

12.7.6 6. Stitched LULC Scenarios

Significance: Combines all allocation zones back into complete Kenya-wide LULC maps for each scenario.

What it does:

1. Stitches together individual allocation blocks
2. Clips to exact Kenya boundary
3. Generates outputs for all scenario combinations (bau, enablers, es_index, LP, FP, LP+FP)

Key Outputs:

```
intermediate/stitched_lulc_simplified_scenarios/  
  lulc_esa_seals7_ssp2_rcp45_luh2-message_bau_2050.tif  
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP_2050.tif  
  lulc_esa_seals7_ssp2_rcp45_luh2-message_FP_2050.tif  
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050.tif  
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050_clipped.tif
```

Significance:

- Provides complete LULC projections
- Enables comparison across scenarios
- Serves as input to ecosystem service models (InVEST)

Kenya LULC Projection: LP+FP Scenario (2050)

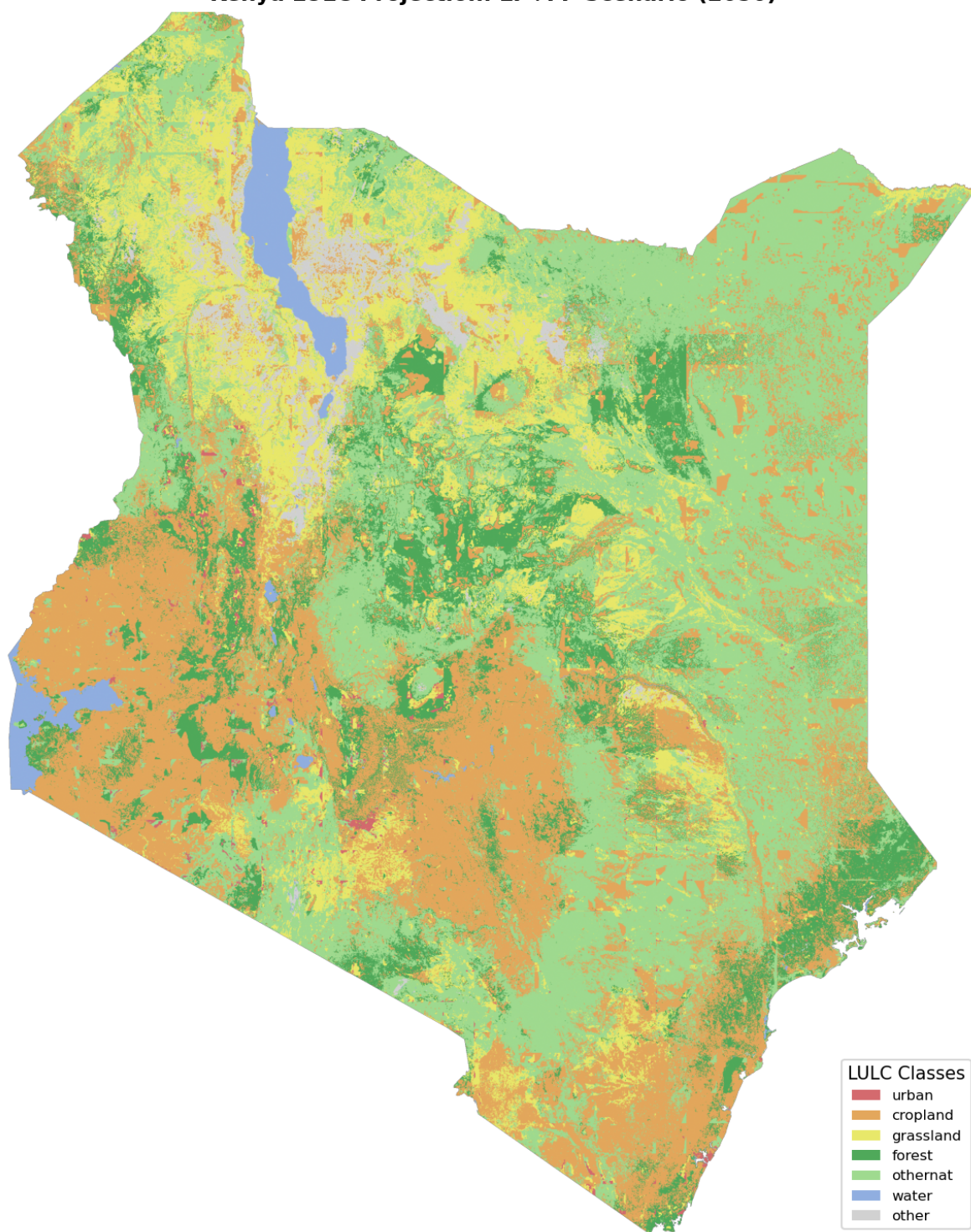


Figure 12.8: Complete LULC projection for Kenya under LP+FP scenario (2050)

12.7.7 7. Visualization

Significance: Creates publication-quality PNG images of LULC projections for easy viewing and sharing.

What it does:

1. Renders each stitched LULC raster as a PNG
2. Applies consistent color schemes
3. Adds appropriate labels and legends

Key Outputs:

```
intermediate/visualization/lulc_pngs/  
  lulc_esa_seals7_ssp2_rcp45_luh2-message_bau_2050.png  
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP_2050.png  
  lulc_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050.png
```

Lulc Ssp2 Rcp45 Luh2-Message Lp+Fp 2050

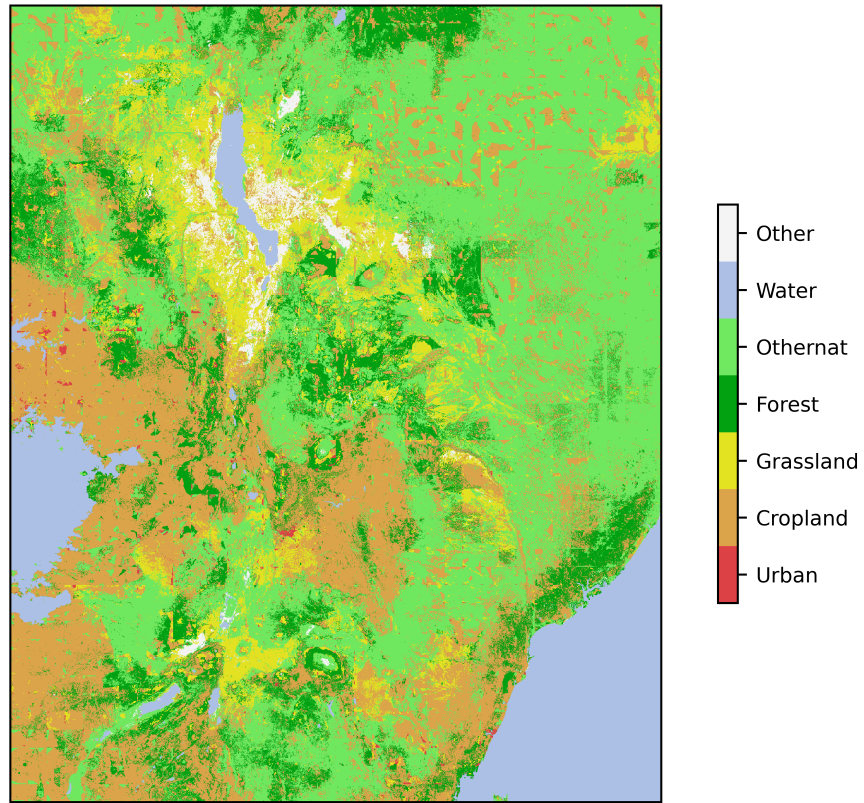


Figure 12.9: Default PNG image for LP+FP scenario

12.7.8 8. InVEST Modules

Significance: Calculates ecosystem service values using Natural Capital Project's InVEST models.

12.7.8.1 8.1 Reproject to Equal Area

Converts LULC maps to equal-area projection (required for accurate area-based calculations).

```
intermediate/reproject_lulc_rasters_to_equal_area/  
lulc_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050.tif
```

12.7.8.2 8.2 Carbon Storage

Estimates above- and below-ground carbon stocks using the InVEST Carbon model.

Key Output:

```
intermediate/run_invest_carbon/  
  esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050/  
    c_storage_bas.tif                # Carbon storage (Mg C/ha)
```

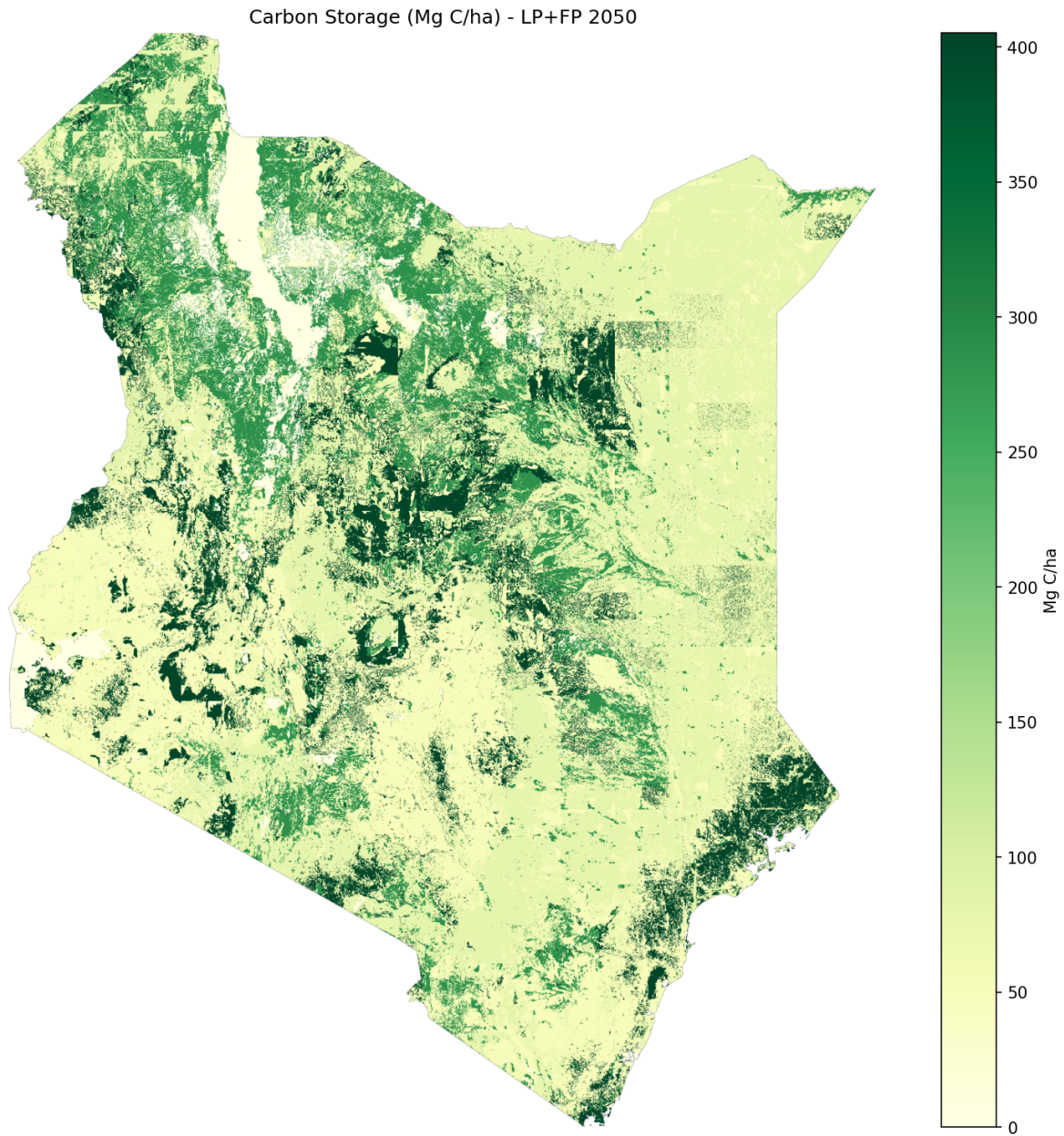



Figure 12.10: Carbon storage under LP+FP scenario (2050)

12.7.8.3 8.3 Pollination

Models pollinator abundance and crop pollination dependency using InVEST's Pollination model.

Key Output:

```
intermediate/run_invest_pollination/  
  esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050/  
    total_pollinator_abundance_summer.tif    # Pollinator index
```

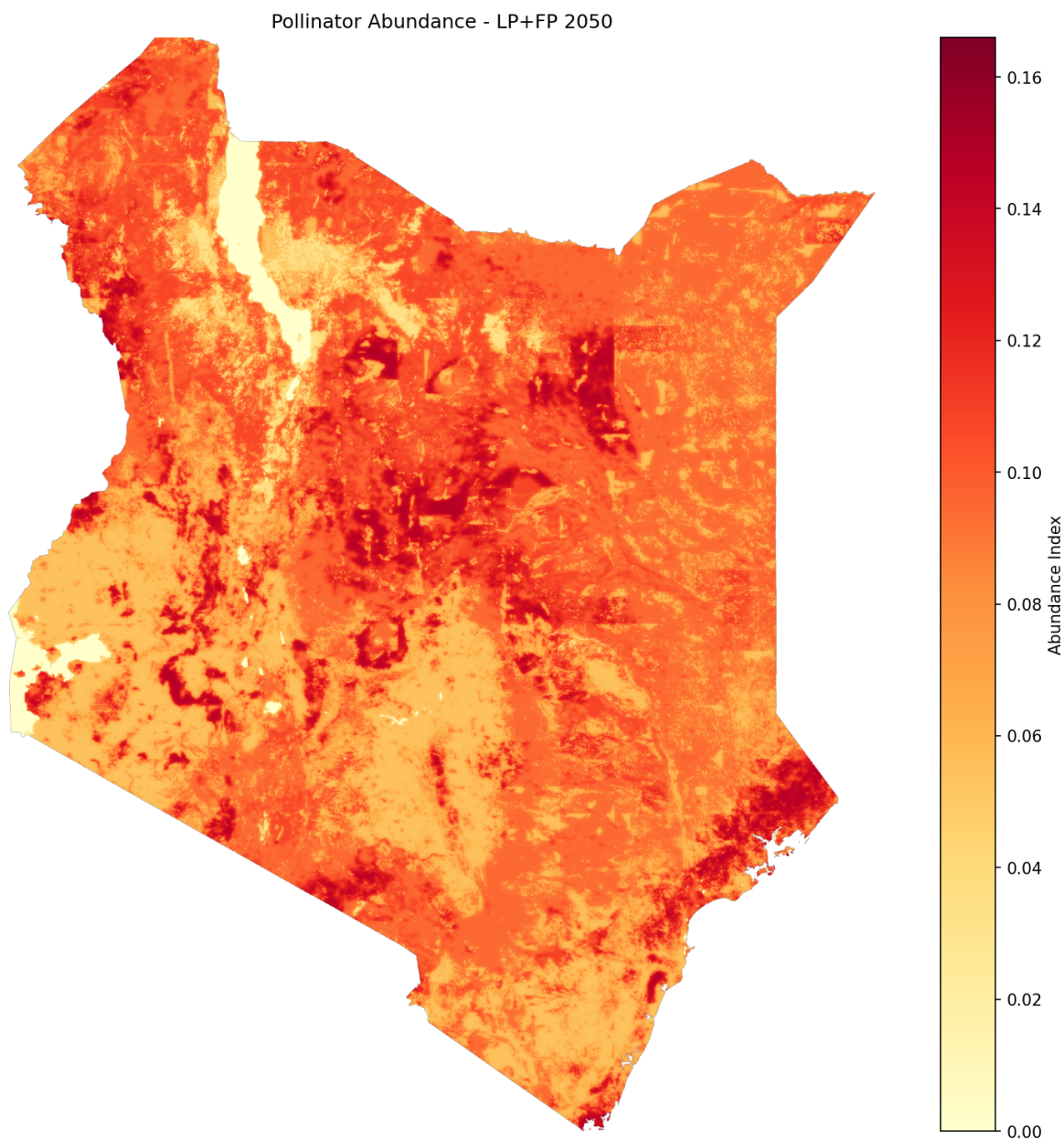


Figure 12.11: Pollinator abundance under LP+FP scenario (2050)

12.7.8.4 8.4 Crop Value and Pollination Shocks

Calculates agricultural value and applies DEMETRA economic shocks based on pollination changes.

What it does:

1. Baseline crop value: Combines production data from all crops weighted by pollination dependency
2. Maximum potential loss: Calculates loss if pollination disappeared completely
3. Pollination sufficiency threshold: Uses 0.3 as the critical threshold
4. Adjusted crop value: For each scenario:
 - If below threshold → apply proportional loss
 - If at or above threshold → maintain baseline value
5. Regional shocks: Calculates $\text{shock_value} = \text{scenario_value} / \text{baseline_value}$ by region

Key Outputs:

```
intermediate/calculate_crop_value_and_shock/  
  crop_value_baseline.tif                # Baseline crop value ($/ha)  
  crop_value_max_lost.tif                # Maximum possible loss ($/ha)  
  crop_value_adjusted_baseline_2020.tif  # Adjusted baseline  
  crop_value_adjusted_esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050.tif # Scenario  
  pollination_shocks_seals7_ssp2_rcp45_luh2-message.csv          # Regional shock values  
  pollination_shocks_alt_seals7_ssp2_rcp45_luh2-message.csv      # Alternative regions
```

Significance:

- Translates ecological changes (pollination) into economic impacts
- Provides spatially explicit economic loss estimates
- Generates shock values for DEMETRA CGE model integration

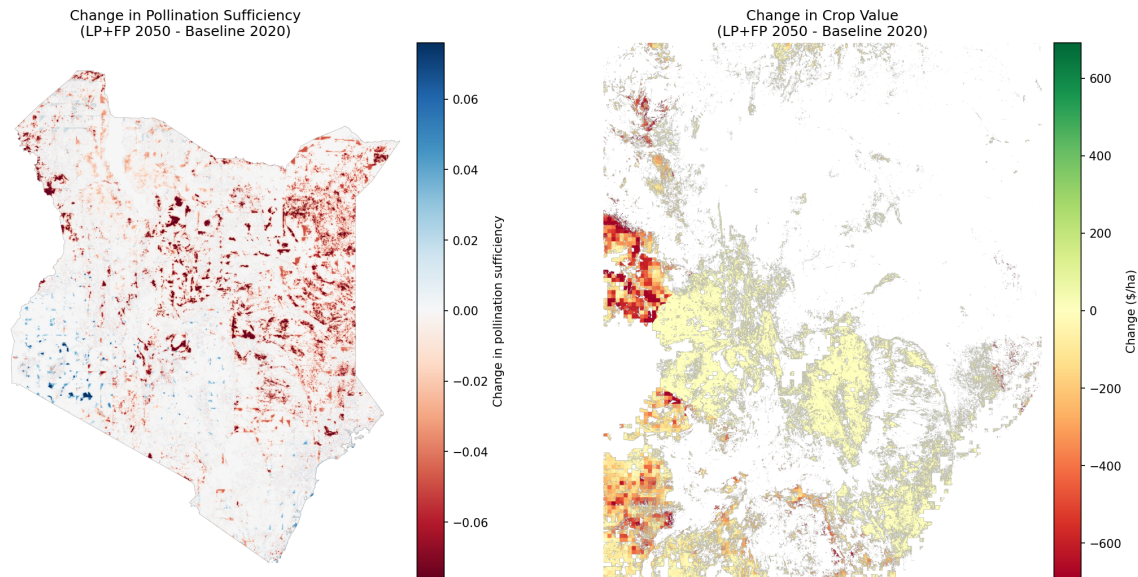


Figure 12.12: Pollination change and Crop value change

Table 12.1: Region Pollination Shocks (BAU, 2050)

Year	Region	Shock_Value
2050	AN	0.86908
2050	AS	0.919913
2050	CO	0.921871
2050	HR	0.924216
2050	MN	0.924557
2050	MO	0.929781
2050	MS	0.977373
2050	NA	0.99532

12.7.8.5 8.5 Biodiversity Index

Calculates habitat quality and biodiversity index based on LULC patterns and biodiversity data (species ranges, endemic species ranges, key biodiversity areas, ecoregions). Not currently an official part of the InVEST ecosystem.

Key Output:

```
intermediate/calculate_biodiversity_index/
esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050/
```


esa_seals7_ssp2_rcp45_luh2-message_LP+FP_2050_index.tif

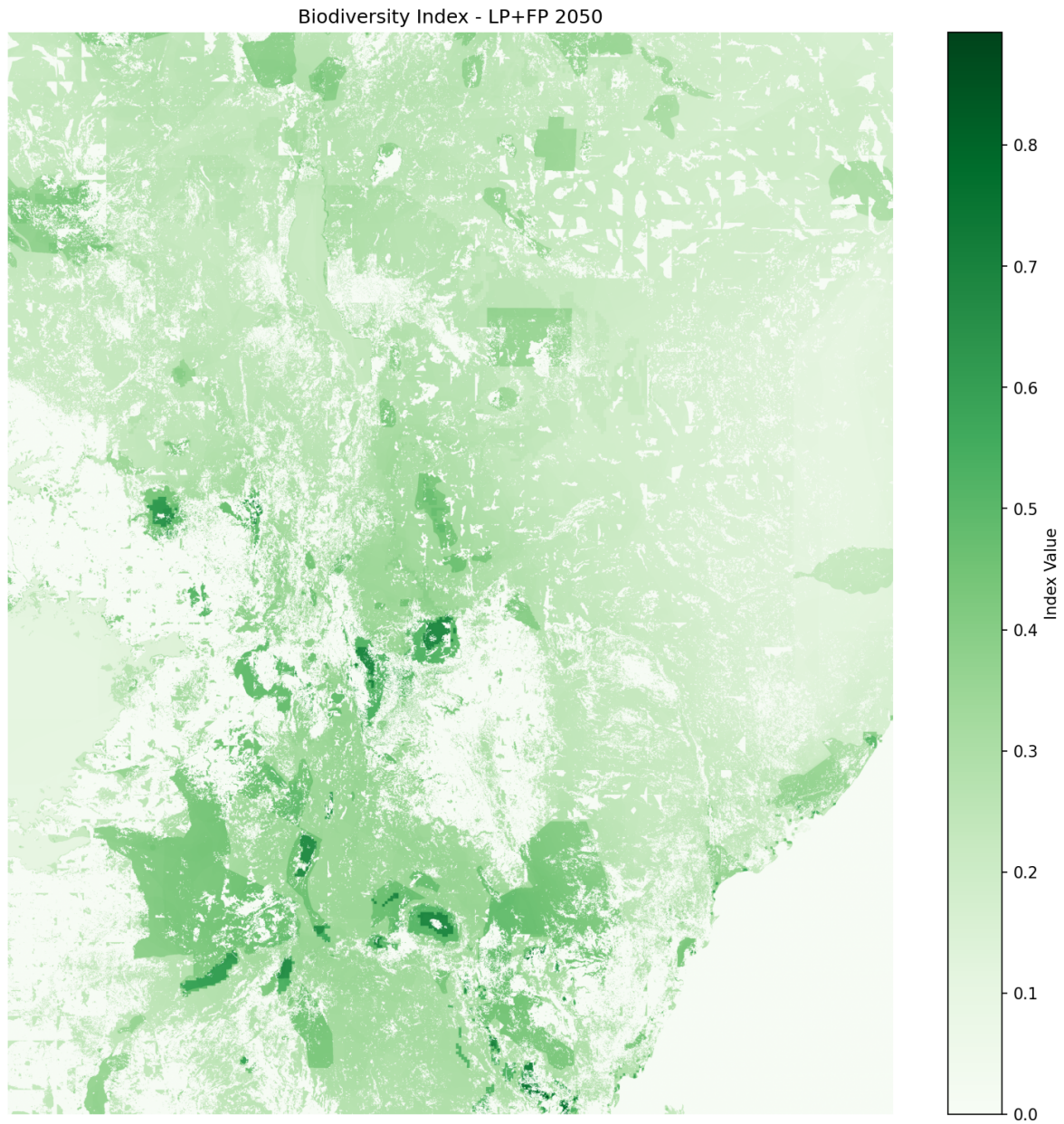


Figure 12.13: Biodiversity index under LP+FP scenario (2050)

12.7.9 9. Summary and Visualization

Significance: Creates multi-panel comparison figures showing results across all scenarios and ecosystem services.

What it does:

1. Compiles results from all InVEST modules
2. Generates comparison figures across scenarios (bau, LP, FP, LP+FP, enablers, es_index)
3. Creates summary visualizations for reports and presentations

Key Outputs:

```
intermediate/summarize_and_visualize_multi_vector/  
  biodiversity_combined/  
    esa_seals7_ssp2_rcp45_luh2-message_bau_biodiversity_combined.png  
  carbon_combined/  
    esa_seals7_ssp2_rcp45_luh2-message_es_index_carbon_combined.png  
  lulc_combined/  
    esa_seals7_ssp2_rcp45_luh2-message_enablers_lulc_combined.png  
  pollination_combined/  
    esa_seals7_ssp2_rcp45_luh2-message_es_index_pollination_combined.png  
  summary_visualizations/  
    summary_grouped_vertical_2050.png          # Master summary figure
```

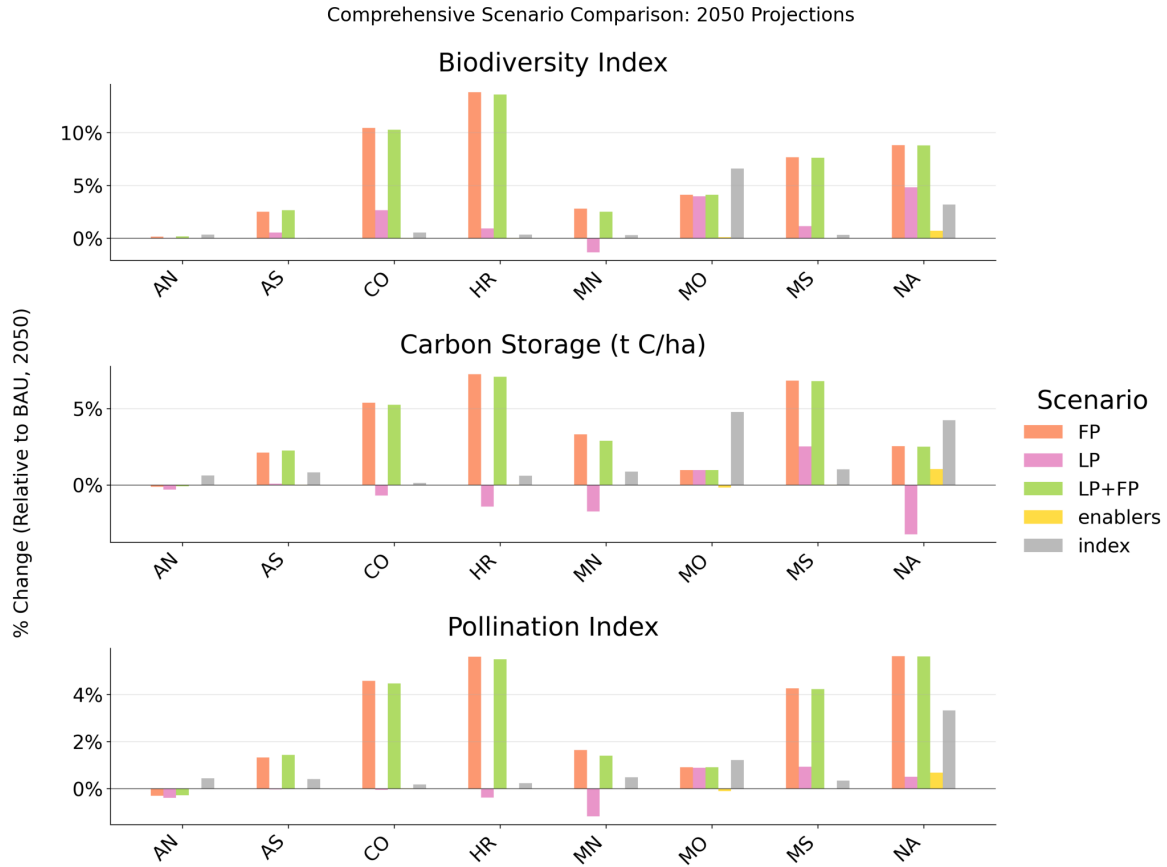


Figure 12.14: Master summary visualization comparing all scenarios (2050)

12.8 Final Outputs

Output files will be generated in:

<home>/Files/seals/projects/run_ken_demetra/

Primary outputs include:

SEALS outputs:

- LULC geotiff files: intermediate/stitched_lulc_simplified_scenarios/

- Land-use/land-cover predictions for each scenario year
- LULC png files: `intermediate/visualization/lulc_pngs/`
 - Spatial representation of land-use transitions

Project outputs:

- Figures: `intermediate/summarize_and_visualize_multi_vector/`
 - `biodiversity_combined`
 - `carbon_combined`
 - `carbon_total_combined`
 - `lulc_combined`
 - `pollination_combined`
 - `summary_visualizations`
-

12.9 Additional Resources

- SEALS Repository: <https://github.com/jandrewjohnson/seals>
- InVEST Documentation: <https://naturalcapitalproject.stanford.edu/software/invest>
- InVEST Python Documentation: <https://invest.readthedocs.io/en/latest/index.html>